

QMC²BO: A Quasi–Monte Carlo Approach to Consensus-Based Optimization and its Convergence to the Minimizer

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Abstract

Consensus-based optimization (CBO) is a powerful gradient-free consensus dynamics for approximating the global minimizer x^* of a possibly non-convex and non-smooth function $f : \mathbb{R}^D \rightarrow \mathbb{R}$. It combines consensus dynamics with stochastic exploration, and its convergence properties has been a very active field of research recently.

In this work, we introduce QMC²BO, a zero-noise variant of CBO, which corresponds to a fully deterministic particle system $x_t = (x_t^1, \dots, x_t^N) \in \mathbb{R}^{D \cdot N}$, $t \geq 0$, with N particles, contraction rate $\lambda > 0$, and concentration parameter $\alpha > 0$. The initial particle configuration x_0 is constructed using low-discrepancy Quasi–Monte Carlo point sets, which provide a uniform coverage of the domain.

Under mild regularity assumptions on f , we show that the particle average $\bar{x}_t^N$ is close to the global minimizer. More precisely, we derive explicit bounds for $\|\bar{x}_t^N - x^*\|$ in terms of the number of particles N and the concentration parameter α , using tools from Quasi–Monte Carlo as well as the Laplace principle. Furthermore, for a given target accuracy $\varepsilon > 0$, we quantify how to choose N , α , λ , and the time T such that $\|\bar{x}_T^N - x^*\| \leq \varepsilon$.

AMS Classification.

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1 Introduction

A central problem in applied mathematics is the global minimization of a potentially non-convex and non-smooth function $f : \mathbb{R}^D \rightarrow \mathbb{R}$, along with the search of its global minimizer $x^* \in \mathbb{R}^D$. Some approaches to this challenge are based on consensus dynamics, where a system of particles explores the energy landscape while continuously adjusting their positions through weighted interactions with one another.

An early application of consensus-based ideas appears in distributed optimization [30, 2], where a global objective is given by the sum of local cost functions $f_n : \mathbb{R}^D \rightarrow \mathbb{R}$, $n = 1, \dots, N$. In this setting, the minimization problem is solved by a system of the form

$$\frac{d}{dt} X_t^n = -\nabla f_n(X_t^n) + \sum_{m=1}^N w_{n,m}(X_t^m - X_t^n), \quad n = 1, \dots, N, \quad (1.1)$$

where $X_t^n \in \mathbb{R}^D$ represents the position of particle n at time t , and the coefficients $w_{n,m} \geq 0$ quantify the interaction strength between particle n and m .

The second term on the right-hand side corresponds to classical consensus dynamics, originally introduced in the context of opinion formation [6, 15]. Under suitable assumptions on the weights $w_{n,m}$, this term drives all agents toward a common state, the consensus [24, 33, 34]. The first term, in contrast, enforces a local gradient descent along each agent’s individual objective function. The interplay between these two components ensures that, under suitable assumptions of convexity and connectivity, the system asymptotically reaches consensus in a minimizer of the aggregate cost function [22, 32, 12].

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More recently, consensus-based optimization (CBO) [25, 4] has been introduced as a gradient-free method to locate global minimizer. In this framework, a set of particles $X_t^1, \dots, X_t^N$ interacts through weighted averages of their positions. In contrast to gradient-based consensus dynamics (1.1), CBO completely eliminates gradient information and instead uses nonlinear, state-dependent interaction weights of the form

$$w_{n,m} = \lambda \frac{\exp(-\alpha f(X_t^m))}{\sum_{k=1}^N \exp(-\alpha f(X_t^k))}. \quad (1.2)$$

This weight choice favors particles in regions, where the objective values is low.

To enhance domain exploration, stochastic perturbations are added, typically of the form

$$\sigma \mathcal{S} \left(\sum_{m=1}^N \frac{w_{n,m}}{\lambda} (X_t^n - X_t^m) \right) dW_t^n, \quad (1.3)$$

where W_t^n is a standard D -dimensional Brownian motion, $\sigma > 0$ controls the noise intensity and $\mathcal{S}$ determines its geometry, typically chosen in isotropic or anisotropic form. The resulting dynamics combine deterministic consensus-driven attraction with stochastic exploration.

We note that many variants of CBO have been proposed in the literature. In this work, we restrict ourselves to the classical formulation described above and do not present the available results in their full generality, in order to keep the presentation concise. A comprehensive and up-to-date survey of CBO methods, extending [29], would be a valuable project in its own right.

The analytical understanding of CBO is well developed. In [4], the authors analyze the mean-field limit of the particle system in the isotropic setting as $N \rightarrow \infty$, leading to the McKean-Vlasov equation

$$d\mathcal{X}_t = -\lambda(\mathcal{X}_t - m_f^\alpha(\mathcal{X}_t)) dt + \sigma |\mathcal{X}_t - m_f^\alpha(\mathcal{X}_t)| dW_t, \quad (1.4)$$

where $m_f^\alpha(\mathcal{X}_t)$ denotes a weighted average with respect to the law of $\mathcal{X}_t$. This equation describes the evolution of the distribution of the particle system in the limit of infinitely many interacting agents.

Under suitable assumptions on f , it is shown in [4] that the solution concentrates exponentially fast around a Dirac measure located arbitrarily close to x^* . A similar result for the anisotropic setting was established in [5].

These convergence results have been extended in [7, 8] under weaker assumptions on f . In particular, they show that for suitable parameter choices, the mean-square distance to the global minimizer decays exponentially up to a finite time horizon, i.e.,

$$\mathbb{E}(\|\mathcal{X}_t - x^*\|^2) \leq \mathbb{E}(\|\mathcal{X}_0 - x^*\|^2) e^{-(1-\vartheta)(2\lambda - \kappa(\mathcal{S})\sigma^2)t}, \quad (1.5)$$

for t in a suitable interval, where $\vartheta \in (0, 1)$ and $\kappa(\mathcal{S})$ is a constant depending on the geometry of the stochastic perturbation. In particular, for appropriate parameter choices, the process reaches an arbitrarily small neighborhood of x^* in finite time.

On the level of finite particle systems, quantitative convergence results have been obtained for the Euler-Maruyama approximation $\hat{X}_{k\Delta t}^i$ of the CBO particle system (1.2) and (1.3), with time step size $0 < \Delta t \leq 1$ and $K \in \mathbb{N}$ time steps. In particular, [3] derives explicit mean-square error bounds. Under suitable assumptions on f and appropriate parameter choices, it is shown that

$$\mathbb{E}(\|\hat{X}_{K\Delta t}^n - x^*\|^2) \leq C_{\text{NA}} \Delta t + \frac{C_{\text{MFA}}}{N} + 2 \mathbb{E}(\|\hat{X}_0^n - x^*\|^2) e^{-(1-\vartheta)(2\lambda - \kappa(\mathcal{S})\sigma^2)K\Delta t}, \quad (1.6)$$

for all $n = 1, \dots, N$.

The error thus decomposes into a time discretization error of order Δt , a mean-field approximation error of order N^{-1} , and an exponentially decaying term reflecting convergence towards the minimizer. Related mean-square convergence results with truncated noise were obtained in [9], while convergence in probability was established in [8].

A central bottleneck in the analysis of CBO has long been a missing propagation of chaos result, which rigorously justifies the mean-field limit. More precisely, such a result quantifies the difference between the finite particle system with interaction weights (1.2) and the limiting mean-field dynamics described by the distribution $\mathcal{X}_t$ in (1.4). This issue arises, since classical results from [21, 28] could not be directly applied due to the non-Lipschitz nature of the interaction weights. It has been resolved in a series of works [17, 11, 8, 20, 1, 10], starting with convergence of the empirical measure to a weak solution of the associated Fokker-Planck equation [17], and leading to uniform-in-time estimates of order $1/N$ for the deviation between particle and mean-field dynamics [10].

An alternative approach to simplify the analysis is to consider modified stochastic dynamics. In particular, if all particles are driven by a common Brownian motion, i.e. $W_t^n = W_t$ for all $n = 1, \dots, N$, one can derive explicit convergence rates for the finite system without relying on propagation of chaos arguments [13, 14]. In this setting, exponential convergence to consensus and quantitative estimates for the distance to the minimizer can be obtained. In contrast, if we consider independent Brownian motion, convergence still holds but with more complex rates depending explicitly on the number of particles N [19].

An alternative approach is to completely remove stochastic perturbation, so the system is purely driven by the consensus dynamics in (1.1) with interaction weights (1.2). In this work, we consider this purely deterministic system

$$\frac{dx_t^n}{dt} = -\lambda (x_t^n - \nu_f^\alpha(x_t)), \quad n = 1, \dots, N, \quad (1.7)$$

where $\lambda > 0$ denotes the interaction strength, $x_t = (x_t^1, \dots, x_t^N) \in \mathbb{R}^{DN}$, and

$$\nu_f^\alpha(x_t) = \sum_{m=1}^N x_t^m \frac{\exp(-\alpha f(x_t^m))}{\sum_{k=1}^N \exp(-\alpha f(x_t^k))} \quad (1.8)$$

is the consensus state at time t . The driving term $x_t^n - \nu_f^\alpha(x_t)$ measures the deviation of the particle n from the current consensus state at time t . Consequently, each particle moves toward a weighted average. Particles with smaller objective function values receive larger weights, so that the consensus state is primarily determined by the positions of particles with smaller function evaluations.

This system can be interpreted as a deterministic variant of classical CBO. Despite the absence of both randomness and gradient information, agents still interact through exponentially weighted averages biased toward lower function values.

This perspective was considered in [cite](#), where a graph-theoretic Lyapunov analysis was used to show that the resulting system is *globally exponentially stable* with rate λ with respect to the set of equilibrium points

$$\mathcal{G}^* = \{x \in \mathbb{R}^{N \cdot D} : x^{1,d} = \dots = x^{N,d}, d = 1, \dots, D\}.$$

More precisely, the evolution of each particle can be expressed explicitly as

$$x_t^n = \bar{x}_t^N + e^{-\lambda t}(x_0^n - \bar{x}_0^N), \quad n = 1, \dots, N, \quad (1.9)$$

where

$$\bar{x}_t^N := \frac{1}{N} \sum_{n=1}^N x_t^n$$

denotes the average of all particles. This result can be seen as the deterministic analogue of (1.5) for the finite- N system. The key differences are that the stochastic perturbation is absent and therefore the exponential convergence rate depends solely on the deterministic drift parameter λ , and that convergence is guaranteed only to the set of equilibrium points $\mathcal{G}^*$, rather than to the global minimizer x^* . In particular, [cite](#) does not specify which point $x^\infty \in \mathcal{G}^*$ is actually approached by the dynamics.

To quantify how close x^∞ is to the global minimizer x^* , the main objective of this work is to analyze the difference between x^* and the average particle position $\bar{x}_t^N$, which converges to x^∞ as $t \rightarrow \infty$. To this end, we introduce the QMC²BO method, a deterministic interacting particle system evolving according to (1.7), complemented by a deterministic choice of initial particle positions. More precisely, the particles are initialized as

$$x_0^n = \mathfrak{s}(\mathbf{p}^n), \quad n = 1, \dots, N,$$

where $\mathbf{p}^1, \dots, \mathbf{p}^N \in [0, 1)^D$ are deterministically chosen points designed to cover the unit cube as uniformly as possible. The mapping $\mathfrak{s}$ rescales these points to $[-K, K)^D$ via

$$\mathfrak{s} : [0, 1]^D \rightarrow [-K, K]^D, \quad \mathfrak{s}(u) = 2Ku - K\mathbf{1}_D, \quad (1.10)$$

for some $K > 0$. This replaces the typically random initialization in classical CBO by deterministic points, allowing us to perform a precise quantitative analysis of the difference between x^* and $\bar{x}_t^N$.

Our main contribution is the derivation of explicit upper bounds for the deviation $\|\bar{x}_t^N - x^*\|$ in terms of the number of particles N and the concentration parameter α , under suitable regularity assumptions

on f . In this sense, our result can be interpreted as a deterministic analogue of (1.6), where discretization errors and expectations are absent, but explicit and fully deterministic rates in N and α are obtained.

To achieve this, we decompose the total error into two contributions: a finite-particle approximation error, arising from the use of a finite number of particles N , and a finite-concentration error, due to $\alpha < \infty$. In Section 2, we first derive this error decomposition. The finite-particle approximation error is analyzed in Subsection 2.1, where we exploit Quasi-Monte Carlo point sets, which provide highly uniform coverings of the unit cube, to obtain explicit convergence rates in N . In Subsection 2.2, we study the finite-concentration error using Laplace principle and derive explicit rates in α . These results are combined in Subsection 2.3 to establish bounds for the deviation $\|\bar{x}_t^N - x^*\|$.

The detailed proofs of the auxiliary results are given in Section 3. In Section 4, we complement our theoretical findings with numerical experiments, illustrating the behavior of the deviation and the sharpness of the derived bounds. Finally, Section 5 discusses and concludes this work.

2 Convergence of the QMC²BO Method to the Minimizer

We aim to understand how this difference between $\bar{x}_t^N$ and x^* depends on the concentration parameter α and on the number of particles N . To make this dependence explicit, we begin by rewriting the system in a form better suited for analysis.

Taking the average of the particle system (1.7) yields the evolution equation

$$\frac{d}{dt}\bar{x}_t^N = \frac{1}{N} \sum_{n=1}^N \frac{d}{dt}x_t^n = \frac{1}{N} \sum_{n=1}^N -\lambda (x_t^n - \nu_f^\alpha(x_t)) = -\lambda (\bar{x}_t^N - \nu_f^\alpha(x_t)).$$

To separate the mean behavior from deviations, define the residual vector

$$r_t = (r_t^1, \dots, r_t^N)^\top = (r_t^{1,1}, \dots, r_t^{1,D}, \dots, r_t^{N,1}, \dots, r_t^{N,D})^\top, \quad (2.1)$$

where $r_t^{n,d} := x_t^{n,d} - \bar{x}_t^{N,d}$ measures the deviation of particle n in dimension d from the particle average. From equation (1.9) we know that the component-wise average $\bar{x}_t^N$ is exponentially attracting, which then implies the compact representation

$$x_t = \begin{pmatrix} \bar{x}_t^N \\ \vdots \\ \bar{x}_t^N \end{pmatrix} + e^{-\lambda t} r_0. \quad (2.2)$$

Substituting this into the mean dynamics yields

$$\frac{d}{dt}\bar{x}_t^N = -\lambda (\bar{x}_t^N - \nu_f^\alpha((\bar{x}_t^N, \dots, \bar{x}_t^N)^\top + e^{-\lambda t} r_0)). \quad (2.3)$$

This shows that the evolution of the D -dimensional average is driven by a perturbed consensus function

A closer inspection of the consensus reveals that

$$\begin{aligned} \nu_f^\alpha((\bar{x}_t^N, \dots, \bar{x}_t^N)^\top + e^{-\lambda t} r_0) &= \sum_{m=1}^N (\bar{x}_t^N + e^{-\lambda t} r_0^m) \frac{\exp(-\alpha f(\bar{x}_t^N + e^{-\lambda t} r_0^m))}{\sum_{k=1}^N \exp(-\alpha f(\bar{x}_t^N + e^{-\lambda t} r_0^k))} \\ &= \bar{x}_t^N + e^{-\lambda t} \sum_{m=1}^N r_0^m \frac{\exp(-\alpha f(\bar{x}_t^N + e^{-\lambda t} r_0^m))}{\sum_{k=1}^N \exp(-\alpha f(\bar{x}_t^N + e^{-\lambda t} r_0^k))} \end{aligned}$$

for $t \geq 0$, using the fact that the weights sum to one. Substituting this into (2.3) yields

$$\frac{d}{dt}\bar{x}_t^N = \lambda e^{-\lambda t} \sum_{m=1}^N r_0^m \frac{\exp(-\alpha f(\bar{x}_t^N + e^{-\lambda t} r_0^m))}{\sum_{k=1}^N \exp(-\alpha f(\bar{x}_t^N + e^{-\lambda t} r_0^k))}.$$

We now isolate how the particle average $\bar{x}_t^N$ approaches the minimizer x^* . Substituting back (2.2) gives

$$\begin{aligned} \frac{d}{dt}\bar{x}_t^N &= \lambda \sum_{m=1}^N (x_t^m - \bar{x}_t^N) \frac{\exp(-\alpha f(x_t^m))}{\sum_{k=1}^N \exp(-\alpha f(x_t^k))} \\ &= \lambda (x^* - \bar{x}_t^N) + \lambda \sum_{m=1}^N (x_t^m - x^*) \frac{\exp(-\alpha f(x_t^m))}{\sum_{k=1}^N \exp(-\alpha f(x_t^k))}, \end{aligned}$$

where we added and subtracted $x^\star$ and used again that the weights sum to one.

Define now

$$F_{N,\alpha}(\bar{x}_t^N, t) := \sum_{m=1}^N (x_t^m - x^\star) \frac{\exp(-\alpha f(x_t^m))}{\sum_{k=1}^N \exp(-\alpha f(x_t^k))} \quad (2.4)$$

and introduce the deviation of the mean from the minimizer $\xi_t := x^\star - \bar{x}_t^N$. Then

$$\frac{d}{dt} \bar{x}_t^N = \lambda(x^\star - \bar{x}_t^N) + F_{N,\alpha}(\bar{x}_t^N, t) = \lambda \xi_t + \lambda F_{N,\alpha}(x^\star - \xi_t, t).$$

The dynamics of ξ_t satisfy

$$\frac{d}{dt} \xi_t = -\frac{d}{dt} \bar{x}_t^N = -\lambda \xi_t - \lambda F_{N,\alpha}(x^\star - \xi_t, t).$$

Multiplying by $e^{\lambda t}$ and rearranging shows that

$$e^{\lambda t} \frac{d}{dt} \xi_t + \lambda e^{\lambda t} \xi_t = -\lambda e^{\lambda t} F_{N,\alpha}(x^\star - \xi_t, t).$$

The left-hand side coincides with the time derivative of the product $e^{\lambda t} \xi_t$. Using this and integrating from 0 to t gives

$$e^{\lambda t} \xi_t - \xi_0 = \int_0^t \frac{d}{ds} (e^{\lambda s} \xi_s) ds = -\lambda \int_0^t e^{\lambda s} F_{N,\alpha}(x^\star - \xi_s, s) ds.$$

Rewriting in terms of $\bar{x}_t^N$ yields the exact representation

$$\bar{x}_t^N = (1 - e^{-\lambda t}) x^\star + e^{-\lambda t} \bar{x}_0^N + \lambda e^{-\lambda t} \int_0^t e^{\lambda s} F_{N,\alpha}(\bar{x}_s^N, s) ds. \quad (2.5)$$

Here,

$$F_{N,\alpha}(z, s) = (z - x^\star) + e^{-\lambda s} \sum_{m=1}^N r_0^m \frac{\exp(-\alpha f(z + e^{-\lambda s} r_0^m))}{\sum_{k=1}^N \exp(-\alpha f(z + e^{-\lambda s} r_0^k))} \quad (2.6)$$

for some $z \in \mathbb{R}^D$.

We see that the particle average is composed of three contributions: the term $(1 - e^{-\lambda t}) x^\star$, converging exponentially in time toward the minimizer; the term $e^{-\lambda t} \bar{x}_0^N$, decaying exponentially fast; and finally the integral contribution, which encodes the dependence on both the concentration parameter α and the number of particles N .

Our aim is now to quantify the distance between the particle average $\bar{x}_t^N$ and the minimizer $x^\star$ in terms of the parameters α and N . To this end, it is essential to analyze the behavior of the term $F_{N,\alpha}(\bar{x}_s^N, s)$. For this reason, for $d = 0, \dots, D$, we define the functions

$$g_d : [-2K, 2K]^D \rightarrow \mathbb{R}, \quad g_d(v) := v_d \exp(-\alpha f(z + e^{-\lambda s} v)), \quad (2.7)$$

where $v = (v_1, \dots, v_D)^\top \in \mathbb{R}^D$ and we set $v_0 = 1$. Furthermore, we introduce

$$I(g_d) := \int_{[0,1]^D} g_d(\mathfrak{s}(x)) dx. \quad (2.8)$$

In Section 2.1, we compare the term

$$\tilde{F}_{N,\alpha}(z, s) := \sum_{m=1}^N r_0^m \frac{\exp(-\alpha f(z + e^{-\lambda s} r_0^m))}{\sum_{k=1}^N \exp(-\alpha f(z + e^{-\lambda s} r_0^k))} \quad (2.9)$$

for $z \in [-K_\rho, K_\rho]^D$ and $s \geq 0$, component-wise for $d = 1, \dots, D$ with the limiting quantity

$$(\tilde{F}_{\infty,\alpha}(z, s))_d := \frac{\int_{[0,1]^D} \mathfrak{s}_d(x) \exp(-\alpha f(z + e^{-\lambda s} \mathfrak{s}(x))) dx}{\int_{[0,1]^D} \exp(-\alpha f(z + e^{-\lambda s} \mathfrak{s}(x))) dx} = \frac{I(g_d)}{I(g_0)}. \quad (2.10)$$

The restriction $z \in [-K, K]^D$ arises from the assumption that all initial points satisfy $x_0^n \in [-K, K]^D$. In particular, Lemma 2.5 will prove

$$\lim_{N \rightarrow \infty} \tilde{F}_{N,\alpha}(z, s) = \tilde{F}_{\infty,\alpha}(z, s). \quad (2.11)$$

using classical tools from Quasi-Monte Carlo theory and assuming that all initial particle position arise from a sequence, which is uniformly distributed modulo one. We will additionally quantify the convergence rate in N . Therefore, $\tilde{F}_{N,\alpha}$ acts as a quadrature approximation of its infinite-particle limit $\tilde{F}_{\infty,\alpha}$.

In Section 2.2 we then analyze the limiting behavior

$$\lim_{\alpha \rightarrow \infty} F_{\infty,\alpha}(z, s) \quad (2.12)$$

for $z \in [-K, K]^D$ and $s \geq 0$ and in Lemma 2.6 we will demonstrate that this term converges to zero, together with an explicit rate in α . The proof will rely on the integrability of polynomials and exponentially decaying weights

$$\int_{\mathbb{R}^D} \|x\|^b \exp(-a\|x\|^\beta) dx = \frac{2}{\beta} \pi^{\frac{D}{2}} a^{-\frac{D+b}{\beta}} \frac{\Gamma((D+b)/\beta)}{\Gamma(D/2)}, \quad (2.13)$$

which follows from a change of variables in polar coordinates, and the following assumptions on the objective function f and its minimizer x^* .

Assumption 2.1. Let $f : \mathbb{R}^D \rightarrow \mathbb{R}$ and $x^* \in \mathbb{R}^D$ be given. We assume the following conditions.

- (A1) **Global minimality and uniqueness.** The point x^* is a unique global minimizer of f . Moreover, for every $\delta > 0$ there exists a constant $c = c(\delta) > 0$ such that

$$f(x) \geq f(x^*) + c$$

for every $x \notin B_\delta^\infty(x^*)$, where

$$B_r^\infty(x) := \{y \in \mathbb{R}^D : \|y - x\|_\infty < r\} \quad (2.14)$$

denotes the open D -dimensional cube with center $x \in \mathbb{R}^D$ and half side-length $r > 0$.

- (A2) **Local growth condition near the minimizer.** There exist constants $\delta_{\text{loc}}, m_f, C_f > 0$ and exponents $\beta, \gamma > 0$ such that

$$f(x) = f(x^*) + m_f \|x - x^*\|^\beta + R(x, x^*)$$

for all $x \in B_{\delta_{\text{loc}}}^\infty(x^*)$, where the remainder term R satisfies

$$|R(x, x^*)| \leq C_f \|x - x^*\|^{\beta+\gamma}.$$

By possibly reducing δ_{loc} , we may further assume that

$$|R(x, x^*)| \leq \frac{m_f}{2} \|x - x^*\|^\beta$$

for all $x \in B_{\delta_{\text{loc}}}^\infty(x^*)$.

- (A3) **Boundedness of the minimizer.** There exists a constant $\rho \in (0, 1)$ such that

$$x^* \in B_{K\rho}^\infty(\mathbf{0}_D), \quad K_\rho := K\rho.$$

Assumption (A1) ensures that $B_{\delta_{\text{loc}}}^\infty(x^*)$ is the unique region in which the global minimum value of f is attained and that f is separated from this minimum outside any neighborhood of x^* . Assumption (A2) provides a quantitative control of the local behavior of f around the minimizer. The growth exponents β and γ determine the sharpness of the minimum. Finally, Assumption (A3) ensures that the minimizer lies in an open cube in the state space from which the initial particle positions $x_0^1, \dots, x_0^N$ are drawn.

The proof of Lemma 2.6, which demonstrates the convergence of $F_{\infty,\alpha}(z, s)$ to $\mathbf{0}_D$ as $\alpha \rightarrow \infty$ will also rely on an application of the Laplace principle of the following form.

Lemma 2.2. Let assumptions 2.1 hold. For any $\delta \leq \delta_{\text{loc}}$, such that $B_\delta^\infty(x^*) \subseteq B_{K e^{-\lambda s}}^\infty(z)$, we have

$$I(g_0) \geq \frac{C_0(\alpha) \alpha^{-\frac{D}{\beta}}}{(2K)^D \exp(\alpha f(x^*))}, \quad (2.15)$$

where

$$C_0(\alpha) := \int_{B_{\delta \alpha^{1/\beta}}^\infty(\mathbf{0}_D)} \exp\left(-m_f \|w\|^\beta - \alpha R\left(x^* + \alpha^{-\frac{1}{\beta}} w, x^*\right)\right) dw \quad (2.16)$$

is positively bounded from below and above and converges to the finite constant

$$\frac{2}{\beta} \pi^{\frac{D}{2}} m_f^{-\frac{D}{\beta}} \frac{\Gamma(D/\beta)}{\Gamma(D/2)} \quad (2.17)$$

as $\alpha \rightarrow \infty$.

2.1 Quasi-Monte Carlo Approximation for the Finite-Particle System

We first recall the notion of low-discrepancy point sets and collect several basic concepts and results that will be used throughout this section. For further information, we refer to Chapter 2 of [23]. Let $\mathcal{P} = \{\mathbf{p}^1, \dots, \mathbf{p}^N\} \subset [0, 1]^D$ be a set of N points. We define

$$A([\mathbf{a}, \mathbf{b}], \mathcal{P}) := \sum_{n=1}^N \mathbb{1}_{[\mathbf{a}, \mathbf{b}]}(\mathbf{p}^n),$$

where $\mathbb{1}_{[\mathbf{a}, \mathbf{b}]}$ denotes the indicator function of the interval $[\mathbf{a}, \mathbf{b}] = \prod_{d=1}^D [a_d, b_d]$. Hence, $A([\mathbf{a}, \mathbf{b}], \mathcal{P})$ counts the number of points of $\mathcal{P}$, which belong to $[\mathbf{a}, \mathbf{b}]$.

This counting function $A([\mathbf{a}, \mathbf{b}], \mathcal{P})$ provides the basis for the notion of *uniform distribution modulo one*, which formalizes the idea that the empirical distribution of a sequence of points converges to the uniform distribution on the unit cube.

Definition 2.3 (Uniform Distribution Modulo One [23, page 14]). A sequence of points $\mathcal{S} = (\mathbf{p}^n)_{n \in \mathbb{N}} \subset [0, 1]^D$ is called *uniformly distributed modulo one*, if

$$\lim_{N \rightarrow \infty} \left| \frac{A([\mathbf{a}, \mathbf{b}], \{\mathbf{p}^1, \dots, \mathbf{p}^N\})}{N} - \lambda_D([\mathbf{a}, \mathbf{b}]) \right| = 0$$

for all $\mathbf{a}, \mathbf{b} \in [0, 1]^D$ with $\mathbf{a} \leq \mathbf{b}$. Here, λ_D denotes the D -dimensional Lebesgue measure.

While Definition 2.3 describes the asymptotic behaviour of an infinite sequence, it is often convenient to quantify how well a finite point set approximates the uniform distribution. This leads to the related notion of discrepancy. For a point set $\mathcal{P} = \{\mathbf{p}^1, \dots, \mathbf{p}^N\} \subset [0, 1]^D$, the *star-discrepancy* is defined as

$$\mathcal{D}_N^*(\mathcal{P}) := \sup_{\mathbf{a} \in [0, 1]^D} \left| \frac{A([\mathbf{0}_D, \mathbf{a}], \mathcal{P})}{N} - \lambda_D([\mathbf{0}_D, \mathbf{a}]) \right|. \quad (2.18)$$

It measures the maximal deviation between the empirical distribution induced by $\mathcal{P}$ and the uniform distribution λ_D , restricted to intervals starting at the origin.

This concept plays a central role in Quasi-Monte Carlo integration. In particular, a sequence $\mathcal{S} = (\mathbf{p}^n)_{n \in \mathbb{N}} \subset [0, 1]^D$ is uniformly distributed modulo one if and only if

$$\lim_{n \rightarrow \infty} \mathcal{D}_N^*(\{\mathbf{p}^1, \dots, \mathbf{p}^N\}) = 0 \quad (2.19)$$

Point sets with small star-discrepancy are well suited for numerical integration. The following theorem summarizes two fundamental discrepancy-based error bounds for Quasi-Monte Carlo integration. The second inequality is the classical *Koksma–Hlawka* inequality.

Theorem 2.4 (Koksma–Hlawka(-type) inequalities [16, 23, Theorem 10.2]). Let $g : [0, 1]^D \rightarrow \mathbb{R}$ be continuous and let $\mathcal{P} = \{\mathbf{p}^1, \dots, \mathbf{p}^N\} \subset [0, 1]^D$ be a set of D -dimensional points. Then the following *Koksma–Hlawka-type* error bound holds:

$$\left| \frac{1}{N} \sum_{i=1}^N g(\mathbf{p}_i) - \int_{[0, 1]^D} g(u) du \right| \leq 4w(g; \mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N)^{1/D}),$$

where

$$w(g; \delta) = \sup_{\substack{u, v \in [0, 1]^D \\ \|u - v\|_\infty \leq \delta}} |g(u) - g(v)|, \quad \delta > 0$$

denotes the modulus of continuity of g . Moreover if g has bounded variation in the sense of Hardy-Krause, then the classical *Koksma–Hlawka* inequality

$$\left| \frac{1}{N} \sum_{i=1}^N g(\mathbf{p}_i) - \int_{[0, 1]^D} g(u) du \right| \leq V(g) \mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N)$$

holds.

Here, $V(g)$ denotes the *Hardy-Krause variation* of g . For sufficiently smooth functions $g \in C^{1,\dots,1}(\mathbb{R}^D; \mathbb{R})$, this quantity is defined as follows: Let $k \in \{1, \dots, D\}$ and let $\mathcal{V} \subseteq \{1, \dots, D\}$ be an index set with $\#\mathcal{V} = k$. For $u \in [0, 1]^k$, we define the point $u_{\mathcal{V}} : \mathbf{1}_{-\mathcal{V}} \in [0, 1]^D$ by

$$(u_{\mathcal{V}} : \mathbf{1}_{-\mathcal{V}})_j = \begin{cases} u_j, & j \in \mathcal{V}, \\ 1, & j \notin \mathcal{V}. \end{cases}$$

Using this notation, we associate to g the k -dimensional function

$$g_{\mathcal{V}}(u) = g(u_{\mathcal{V}} : \mathbf{1}_{-\mathcal{V}}), \quad u \in [0, 1]^k.$$

For a function $f : [0, 1]^k \rightarrow \mathbb{R}$ that is k times continuously differentiable, we define its variation by

$$V(f) = \int_{[0, 1]^k} \left| \frac{\partial^k}{\partial s_1 \cdots \partial s_k} f(s_1, \dots, s_k) \right| d(s_1, \dots, s_k),$$

given that the right-hand side exists. With these definitions, the Hardy-Krause variation of g is given by

$$V(g) = \sum_{k=1}^D \sum_{\substack{\mathcal{V} \subseteq \{1, \dots, D\} \\ \#\mathcal{V}=k}} V(g_{\mathcal{V}}).$$

Using these tools, we can now establish the following lemma, which relates $\tilde{F}_{N,\alpha}(z, s)$ and $\tilde{F}_{\infty,\alpha}(z, s)$.

Lemma 2.5. Let assumptions 2.1 hold and let $f : \mathbb{R}^D \rightarrow \mathbb{R}$ be locally Lipschitz continuous and differentiable. Fix $z \in [-K, K]^D$, $\alpha \geq 1$, $\lambda > 0$ and $s \geq 0$. Assume that the sequence of points $\mathcal{S} = (\mathbf{p}^n)_{n \in \mathbb{N}} \subset [0, 1]^D$ is uniformly distributed modulo one and that there exists $\delta \leq \delta_{\text{loc}}$, such that $B_{\delta}^{\infty}(x^*) \subseteq B_{Ke^{-\lambda s}}^{\infty}(z)$. Moreover, let $\tilde{F}_{N,\alpha}(z, s)$ and $\tilde{F}_{\infty,\alpha}(z, s)$ be given by (2.9) and (2.10), respectively. Then

$$\lim_{N \rightarrow \infty} \tilde{F}_{N,\alpha}(z, s) = \tilde{F}_{\infty,\alpha}(z, s).$$

More precisely, the component-wise approximation error admits the quantitative bound

$$\left| (\tilde{F}_{N,\alpha}(z, s) - \tilde{F}_{\infty,\alpha}(z, s))_d \right| \leq C_{f,K,D} (\mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N))^{1/D} \frac{\alpha^{\frac{D}{\beta}+1}}{C_0(\alpha)}$$

for any $d \in \{1, \dots, D\}$ where the constant $C_{f,K,D}$ depends solely on f , K and D . The term $C_0(\alpha)$ is defined in (2.16), satisfies the bounds in (3.1), and converges to the finite constant provided in (2.17).

The proof of Lemma 2.5 can be found in Section 3. The estimate in Lemma 2.5 shows that the dependence on the number of points N acts only through the star discrepancy $\mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N)$. As the sequence of points $\mathcal{S}$ is uniformly distributed modulo one, this quantity converges to zero as $N \rightarrow \infty$. The convergence rate becomes slower with increasing dimension D , reflecting the well-known curse of dimensionality for deterministic quadrature rules for C^1 integrands. ref As a possible outlook, stronger regularity assumptions on f allow for sharper estimates via the Koksma-Hlawka inequality, avoiding the $1/D$ -dependence in the exponent of the star discrepancy.

The concentration parameter α enters the bound through $\alpha^{\frac{D}{\beta}+1}/C_0(\alpha)$. Here, the factor $\alpha^{\frac{D}{\beta}}$ originates from the Laplace principle in Lemma 2.2, while the additional factor α arises from the Lipschitz constant of g_d , more precisely from its gradient. All remaining constants depend only on the dimension D , K and the regularity of f through $\|\nabla f\|_{\infty}$.

2.2 Laplace Principle for the Finite-Concentration System

We now analyze the finite-concentration system by investigating the asymptotic behavior of $F_{\infty,\alpha}(z, s)$ as $\alpha \rightarrow \infty$. Our approach is based on the Laplace principle from Lemma 2.2, which allows us to rigorously quantify how integrals weighted by exponentially concentrated densities localize around the global minimizer of the objective function. We first recall the relation between the finite-particle system and its limit $N \rightarrow \infty$. Inserting (2.9) into (2.6) and applying Lemma 2.5, the limit of $F_{N,\alpha}(z, s)$ as $N \rightarrow \infty$ is given by

$$F_{\infty,\alpha}(z, s) := \lim_{N \rightarrow \infty} F_{N,\alpha}(z, s) = (z - x^*) + e^{-\lambda s} \tilde{F}_{\infty,\alpha}(z, s) \quad (2.20)$$

for all $z \in [-K, K]^D$ and $s \geq 0$, where $\tilde{F}_{\infty,\alpha}(z, s)$ is defined in (2.10). In order to establish convergence of the finite system, it suffices to analyze the asymptotic behavior of $F_{\infty,\alpha}(z, s)$ as $\alpha \rightarrow \infty$.

Lemma 2.6. Suppose that Assumptions 2.1 hold, and let $F_{\infty,\alpha}(z, s)$ be defined by (2.20). Assume that there exists a radius $\delta \leq \delta_{\text{loc}}$ such that $B_\delta^\infty(x^*) \subseteq B_{Ke^{-\lambda s}}^\infty(z)$ for some arbitrary $z \in [-K, K]^D$ and $s \geq 0$. Then

$$\lim_{\alpha \rightarrow \infty} F_{\infty,\alpha}(z, s) = \mathbf{0}_D.$$

More precisely, the component-wise approximation error admits the quantitative bound

$$|(F_{\infty,\alpha}(z, s))_d| \leq \frac{\alpha^{-\frac{1+\gamma}{\beta}}}{C_0(\alpha)} C_{f,D,\beta,\gamma} + \frac{\alpha^{\frac{D}{\beta}}}{C_0(\alpha) \exp(\alpha c)} (2K)^{D+1} e^{-\lambda s(D+1)},$$

where $C_{f,D,\beta,\gamma}$ is a constant depending only on f, D, β and γ . Moreover, the normalization factor $C_0(\alpha)$ converges, as $\alpha \rightarrow \infty$, to a positive constant depending only on f, D and β , but not on α .

The proof of Lemma 2.6 can be found in Section 3. The estimate in Lemma 2.6 shows that the asymptotic behaviour of $F_{\infty,\alpha}(z, s)$ is bounded by two different contributions. The first term decays polynomially in α at rate $\alpha^{-(1+\gamma)/\beta}$ and reflects the local approximation of the objective function f in a neighborhood of the minimizer x^* .

The second term is exponentially small in α and captures the contribution from outside the local neighbourhood of x^* . Its decay rate is dominated by the factor $\exp(-\alpha c)$, while the remaining prefactor depends only polynomially on α and on the dimension through K^{D+1} . Hence, for sufficiently large α , this term becomes very small.

The normalization factor $C_0(\alpha)$ converges to a positive constant as $\alpha \rightarrow \infty$ and therefore does not affect the asymptotic convergence rate. Overall, the lemma establishes that $F_{\infty,\alpha}(z, s)$ converges to zero as $\alpha \rightarrow \infty$, with a convergence rate that increases with increasing dimension D .

2.3 Combined Convergence Analysis of QMC²BO

In this section, we combine the Quasi-Monte Carlo approximation from Lemma 2.5 with the Laplace principle estimate from Lemma 2.6 in order to derive the convergence of the average particle position $\bar{x}_t^N$ of the QMC²BO dynamics towards the minimizer x^* of the objective function f . In particular, we will obtain an explicit convergence rate that depends on both the number of particles N and the concentration parameter α .

According to (2.5), this difference admits the representation

$$\bar{x}_t^N - x^* = e^{-\lambda t} (\bar{x}_0^N - x^*) + \lambda e^{-\lambda t} \int_0^t e^{\lambda s} F_{N,\alpha}(\bar{x}_s^N, s) ds. \quad (2.21)$$

Here, $F_{N,\alpha}(\bar{x}_s^N, s)$ is defined as in (2.6). Consequently, controlling the convergence behavior of $\bar{x}_t^N - x^*$ reduces to obtaining suitable bounds on this integral term. Using (2.6) and (2.9), we can rewrite

$$\begin{aligned} |(F_{N,\alpha}(\bar{x}_s^N, s))_d| &= |(\bar{x}_s^N - x^*)_d + e^{-\lambda s} (\tilde{F}_{\infty,\alpha}(\bar{x}_s^N, s))_d + e^{-\lambda s} (\tilde{F}_{N,\alpha}(\bar{x}_s^N, s) - \tilde{F}_{\infty,\alpha}(\bar{x}_s^N, s))_d| \\ &= |(F_{\infty,\alpha}(\bar{x}_s^N, s))_d + e^{-\lambda s} (\tilde{F}_{N,\alpha}(\bar{x}_s^N, s) - \tilde{F}_{\infty,\alpha}(\bar{x}_s^N, s))_d|, \end{aligned}$$

where, in the last step, we employ identity (2.20). This decomposition separates the Quasi-Monte Carlo approximation error from the error that can be bounded by the Laplace principle.

Under the assumptions of Lemma 2.5, the Quasi-Monte Carlo error can be bounded by

$$|(\tilde{F}_{N,\alpha}(\bar{x}_s^N, s) - \tilde{F}_{\infty,\alpha}(\bar{x}_s^N, s))_d| \leq C_{\text{QMC}}(N, \alpha)$$

for each component $d = 1, \dots, D$, where

$$C_{\text{QMC}}(N, \alpha) := C_{f,K,D}(\mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N))^{1/D} \frac{\alpha^{\frac{D}{\beta}+1}}{C_0(\alpha)}. \quad (2.22)$$

Here, the constant $C_{f,K,D}$ depends solely on f, K and D and the term $C_0(\alpha)$ is defined in (2.16), satisfies the bounds in (3.1), and converges to the finite constant provided in (2.17).

Assume in addition that the assumptions of Lemma 2.6 are satisfied. Then

$$|(F_{\infty,\alpha}(\bar{x}_s^N, s))_d| \leq C_{\text{Lap}}(\alpha),$$

where the constant $\mathbf{C}_{\text{Lap}}(\alpha)$ is defined by

$$\mathbf{C}_{\text{Lap}}(\alpha) := \frac{\alpha^{-\frac{1+\gamma}{\beta}}}{C_0(\alpha)} C_{f,D,\beta,\gamma} + \frac{\alpha^{\frac{D}{\beta}}}{C_0(\alpha) \exp(\alpha c)} (2K)^{D+1}. \quad (2.23)$$

Combining this bound with the Quasi-Monte Carlo estimate and using the fact that the exponential function is bounded by one for non-positive arguments, we obtain the component-wise estimate

$$|(F_{N,\alpha}(\bar{x}_s^N, s))_d| \leq \mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha). \quad (2.24)$$

Integrating this estimate over time yields

$$\int_0^t e^{\lambda s} |(F_{N,\alpha}(\bar{x}_s^N, s))_d| ds \leq \frac{\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)}{\lambda} (e^{\lambda t} - 1).$$

Substituting this estimate into equation (2.21) allows us to bound each component of the deviation from the minimizer from above by

$$(\bar{x}_s^N - x^*)_d \leq e^{-\lambda s} (\bar{x}_0^N - x^*)_d + (\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)) (1 - e^{-\lambda s}) \quad (2.25)$$

and, analogously, from below by

$$(\bar{x}_s^N - x^*)_d \geq e^{-\lambda s} (\bar{x}_0^N - x^*)_d - (\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)) (1 - e^{-\lambda s}). \quad (2.26)$$

The above bounds form the basis for a quantitative convergence analysis of the average particle position $\bar{x}_s^N$ of the QMC²BO dynamics towards the minimizer x^* . In particular, for a fixed time horizon $[0, T]$, they allow us to verify the existence of a radius $0 < \delta \leq \delta_{\text{loc}}$ such that

$$B_\delta^\infty(x^*) \subseteq \bar{B}_{Ke^{-\lambda s}}^\infty(\bar{x}_s^N),$$

holds for all $s \in [0, T]$, which is a necessary condition required for the application of Lemma 2.6.

Before stating and proving the final result in Theorem 2.8, we briefly summarize the overall setting underlying the QMC²BO dynamics.

Method 2.7. [QMC²BO: Quasi-Monte Carlo Consensus-Based Optimization] Let $N = 2^m$, $m \in \mathbb{N}$, denote the number of particles in dimension D and let $[0, T]$ be a fixed time horizon. The QMC²BO method is defined as a deterministic interacting particle system $x_t^1, \dots, x_t^N$, $t \in [0, T]$, evolving according to

$$\frac{d}{dt} x_t^n = -\lambda (x_t^n - \nu_f^\alpha(x_t)), \quad n = 1, \dots, N,$$

where $\lambda > 0$ and $\alpha \geq 1$ are fixed parameters and the consensus point $\nu_f^\alpha(x_t)$ is defined by (1.8). The initial particle positions are given by $x_0^n = \mathfrak{s}(\mathbf{p}^n)$, where the point sequence $\mathcal{S} = (\mathbf{p}^n)_{n \in \mathbb{N}} \subset [0, 1]^D$ is uniformly distributed modulo one and the linear mapping $\mathfrak{s}$, defined in (1.10), maps the unit cube $[0, 1]^D$ onto the bounded domain $[-K, K]^D$ for some $K \in \mathbb{R}$.

For the QMC²BO method described in Method 2.7, we now establish a quantitative convergence result for the average particle position. Under additional regularity assumptions on the objective function f and for sufficiently large values of the concentration parameter α , we show that the average particle converges to the global minimizer of f , with an explicit error bound depending on N and α .

Theorem 2.8. Consider the QMC²BO method described in Method 2.7. Let $f : \mathbb{R}^D \rightarrow \mathbb{R}$ be locally Lipschitz continuous, differentiable and satisfy Assumptions 2.1 with radius

$$\delta = \min \left(\delta_{\text{loc}}, e^{-\lambda T} \left(\frac{K(1-\rho)}{2} - \max_{d=1, \dots, D} |(\bar{x}_0^N)_d| \right) \right).$$

Furthermore, let $\alpha > 0$ be such that

$$(e^{\lambda T} - 1) (\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)) \leq \frac{K(1-\rho)}{2},$$

where the constants $\mathbf{C}_{\text{QMC}}(N, \alpha)$ and $\mathbf{C}_{\text{Lap}}(\alpha)$ are defined in (2.22) and (2.23), respectively.

Then, for any $t \in [0, T]$, the deviation of the average particle from the minimizer admits the bound

$$\|\bar{x}_t^N - x^*\| \leq e^{-\lambda t} \|\bar{x}_0^N - x^*\| + \frac{\sqrt{D}}{\lambda} (\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)) (1 - e^{-\lambda t}),$$

which explicitly quantifies the contributions of the Quasi-Monte Carlo approximation and the Laplace principle error.

Proof. The main idea of the proof is to apply Lemma 2.5 and Lemma 2.6 with the choice $z = \bar{x}_s^N$. Since the initial particle positions $x_0^1, \dots, x_0^N$ lie in the bounded domain $[-K, K]^D$, their average $\bar{x}_0^N$ belongs to $[-K, K]^D$ as well. Moreover, the convex hull of the particle system contracts over time, implying that $\bar{x}_s^N \in [-K, K]^D$ for all $s \in [0, T]$. As a consequence, the assumptions of Lemma 2.5 and Lemma 2.6 are satisfied, if we verify that our choice of δ, N and α ensures the inclusion

$$B_\delta^\infty(x^*) \subseteq B_{Ke^{-\lambda s}}^\infty(\bar{x}_s^N)$$

for any $s \in [0, T]$. Geometrically, this condition is equivalent to requiring that the cube centered at x^* with side length 2δ is contained in the cube centered at $\bar{x}_s^N$ with side length $2Ke^{-\lambda s}$. Since the cube is symmetric, this inclusion holds if and only if

$$(x^*)_d + \delta \leq (\bar{x}_s^N)_d + Ke^{-\lambda s}, \quad (\bar{x}_s^N)_d - Ke^{-\lambda s} \leq (x^*)_d - \delta$$

for any $d \in \{1, \dots, D\}$.

We therefore fix an arbitrary component $d \in \{1, \dots, D\}$ and establish these inequalities under the assumptions imposed on δ, N , and α . We begin with the initial time $s = 0$. By the assumption on δ , we have

$$\delta \leq e^{-\lambda T} \left(\frac{K(1-\rho)}{2} - \max_{d=1, \dots, D} |(\bar{x}_0^N)_d| \right) < K(1-\rho) - \max_{d=1, \dots, D} |(\bar{x}_0^N)_d|,$$

which immediately yields

$$(\bar{x}_0^N)_d \leq K(1-\rho) - \delta, \quad -(\bar{x}_0^N)_d \leq K(1-\rho) - \delta.$$

Furthermore, Assumption (A3) implies that $(x^*)_d \in (-K\rho, K\rho)$, and hence

$$(\bar{x}_0^N)_d - K \leq (x^*)_d - \delta, \quad (\bar{x}_0^N)_d + K \geq (x^*)_d + \delta$$

follows directly. Since the component d was chosen arbitrarily, we conclude that $B_\delta^\infty(x^*) \subseteq B_K^\infty(\bar{x}_0^N)$.

We now turn to the case $s > 0$. To secure the desired inclusion in the d -th component, we introduce the quantities

$$\chi_\pm : [0, \infty) \rightarrow \mathbb{R}, \quad \chi_\pm(s) := \pm(x^* - \bar{x}_s^N)_d + \delta - Ke^{-\lambda s}.$$

The inclusion $B_\delta^\infty(x^*) \subseteq B_{Ke^{-\lambda s}}^\infty(\bar{x}_s^N)$ is guaranteed in the d -th component whenever $\chi_\pm(s) \leq 0$. Motivated by this observation, we define the stopping time

$$t_\pm^* := \sup\{t \geq 0 : \chi_\pm(s) \leq 0 \text{ for any } s \in [0, t]\}.$$

Since the inclusion has already been verified at $s = 0$, this stopping time is well-defined and satisfy $t_\pm^* \geq 0$.

For every $s \in [0, t_\pm^*]$, the inclusion is true. Consequently, all assumptions of Lemma 2.5 and Lemma 2.6 are satisfied on this interval. We may therefore apply the component-wise estimates (2.25) and (2.26), which yield

$$\chi_\pm(s) = \pm(x^* - \bar{x}_s^N)_d + \delta - Ke^{-\lambda s} \leq \delta + C - (C + A_\pm)e^{-\lambda s}$$

where we have used $C := \mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)$ and $A_\pm := K \pm (\bar{x}_0^N - x^*)_d$.

To determine for which times s the condition $\chi_\pm(s) \leq 0$ is true, we define the auxiliary function

$$H : [0, \infty) \rightarrow \mathbb{R}, \quad H(s) = \delta + C - (C + A_\pm)e^{-\lambda s}$$

whose derivative is given by

$$H'(s) = \lambda(C + A_\pm)e^{-\lambda s}.$$

Since $H(s) \leq 0$ directly implies $\chi_\pm(s) \leq 0$, it suffices to establish negativity of H on $[0, T]$. We distinguish three cases according to the sign of $C + A_\pm$.

Case 1: $C + A_\pm > 0$. In this case, H is strictly monotonically increasing. Using Assumption (A3), the condition imposed on δ , and the assumption on α , we obtain

$$\begin{aligned} H(T) &= \delta + C - (C + K \pm (\bar{x}_0^N - x^*)_d)e^{-\lambda T} \\ &< \delta + C - \left(C + K - \max_{d=1, \dots, D} |(\bar{x}_0^N)_d| - K\rho \right) e^{-\lambda T} \\ &\leq C(1 - e^{-\lambda T}) + e^{-\lambda T} \left(\frac{K(1-\rho)}{2} - K(1-\rho) \right) \\ &\leq e^{-\lambda T} \left(\frac{K(1-\rho)}{2} + \frac{K(1-\rho)}{2} - K(1-\rho) \right) \\ &= 0. \end{aligned}$$

Hence $H(T) < 0$, and monotonicity implies that $H(s) < 0$ for all $s \in [0, T]$.

Case 2: $C + A_{\pm} < 0$. In this case, H is strictly monotonically decreasing. Using Assumption (A3) together with the condition on δ , we derive

$$\begin{aligned} H(0) &= \delta + C - (C + K \pm (\bar{x}_0^N - x^*)_d) \\ &< K(1 - \rho) - \max_{d=1, \dots, D} |(\bar{x}_0^N)_d| - K + \max_{d=1, \dots, D} |(\bar{x}_0^N)_d| + K\rho \\ &= 0. \end{aligned}$$

Therefore $H(0) < 0$, and monotonicity again yields $H(s) < 0$ for all $s \in [0, T]$.

Case 3: $C + A_{\pm} = 0$. Here the function H is constant in time. Using Assumption (A3), the condition on δ , and the identity $C = -A_{\pm}$, we obtain

$$\begin{aligned} H(T) &= \delta + C \\ &< K(1 - \rho) - \max_{d=1, \dots, D} |(\bar{x}_0^N)_d| - K \mp (\bar{x}_0^N - x^*)_d \\ &\leq K\rho - \max_{d=1, \dots, D} |(\bar{x}_0^N)_d| + \max_{d=1, \dots, D} |(\bar{x}_0^N)_d| + K\rho \\ &= 0. \end{aligned}$$

Thus $H(s) < 0$ for all $s \in [0, T]$.

We now show that the stopping time $t_{\pm}^*$ cannot occur before T . Assume, for contradiction, that $t_{\pm}^* < T$. By continuity of $\chi_{\pm}$ and the definition of $t_{\pm}^*$, we must have

$$0 = \chi_{\pm}(t_{\pm}^*) \leq H(t_{\pm}^*) < 0.$$

The first inequality follows because the inclusion is true up to time $t_{\pm}^*$, allowing us to apply the bounds (2.25) and (2.26). The final strict inequality follows from the fact that $H(s) < 0$ for all $s \in [0, T]$ in each of the three cases considered above. This contradiction shows that $t_{\pm}^* \geq T$.

Since the component d was arbitrary, we conclude that the inclusion $B_{\delta}^{\infty}(x^*) \subseteq B_{Ke^{-\lambda s}}^{\infty}(\bar{x}_s^N)$ holds for every $s \in [0, T]$. Consequently, all assumptions of Lemma 2.5 and Lemma 2.6 are verified throughout the entire time interval. Starting from the representation (??) and applying the triangle inequality, we obtain

$$\begin{aligned} \|\bar{x}_t^N - x^*\| &= \left\| e^{-\lambda t}(\bar{x}_0^N - x^*) + e^{-\lambda t} \int_0^t e^{\lambda s} F_{N, \alpha}(\bar{x}_s^N, s) ds \right\| \\ &\leq e^{-\lambda t} \|\bar{x}_0^N - x^*\| + e^{-\lambda t} \int_0^t e^{\lambda s} \|F_{N, \alpha}(\bar{x}_s^N, s)\| ds. \end{aligned}$$

Using the definition of the Euclidean norm together with the component-wise estimate (2.24), we further obtain

$$\|F_{N, \alpha}(\bar{x}_s^N, s)\| = \sqrt{\sum_{d=1}^D |(F_{N, \alpha}(\bar{x}_s^N, s))_d|^2} \leq \sqrt{D} (C_{\text{Lap}}(\alpha) + C_{\text{QMC}}(N, \alpha)).$$

Integrating this bound over time leads to

$$\int_0^t e^{\lambda s} \|F_{N, \alpha}(\bar{x}_s^N, s)\| ds \leq \sqrt{D} (C_{\text{Lap}}(\alpha) + C_{\text{QMC}}(N, \alpha)) \frac{1}{\lambda} (e^{\lambda t} - 1).$$

Combining the above estimates, we finally arrive at

$$\|\bar{x}_t^N - x^*\| \leq e^{-\lambda t} \|\bar{x}_0^N - x^*\| + \frac{\sqrt{D}}{\lambda} (C_{\text{Lap}}(\alpha) + C_{\text{QMC}}(N, \alpha)) (1 - e^{-\lambda t}),$$

which establishes the claimed deviation of the average particle from the minimizer, with an explicit dependence on N and α . $\square$

Theorem 2.8 provides the final quantitative bound for the deviation of the average particle $\bar{x}_t^N$ from the global minimizer x^* . The first term reflects the exponential contraction of the average dynamics with rate λ , while the second term captures the finite approximation error, separating the contributions arising from the Quasi-Monte Carlo point sets and the Laplace principle.

More precisely, this approximation error is controlled by the sum $\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)$, whose structure has been analyzed in Lemma 2.6 and Lemma 2.5. As discussed there, the Laplace contribution $\mathbf{C}_{\text{Lap}}(\alpha)$ decays with increasing α , while the QMC contribution $\mathbf{C}_{\text{QMC}}(N, \alpha)$ is governed by the star discrepancy and thus vanishes as $N \rightarrow \infty$. Consequently, this estimate provides a guideline for parameter selection: for a prescribed accuracy $\varepsilon > 0$, one can quantify how to choose N , α , as well as K , λ , and the time horizon T , such that $\|\bar{x}_T^N - x^*\| \leq \varepsilon$.

In the following two remarks, we discuss the precise role and implications of the requirements on δ and α in Theorem 2.8.

Remark 2.9. In order to secure that the δ from Theorem 2.8 is positive, we have to ensure

$$\frac{K(1-\rho)}{2} \geq \max_{d=1,\dots,D} |(\bar{x}_0^N)_d|. \quad (2.27)$$

Since the integral of $2x - 1$ over $[0, 1]^D$ is zero, we can exploit the construction of the initial particle average $\bar{x}_0^N$ to compute

$$|(\bar{x}_0^N)_d| = K \left| \frac{1}{N} \sum_{n=1}^N 2(\mathbf{p}^n)_d - 1 \right| = K \left| \int_{[0,1]^D} 2x - 1 \, dx - \frac{1}{N} \sum_{n=1}^N 2(\mathbf{p}^n)_d - 1 \right| \leq 2K\mathcal{D}_N^*(\mathbf{p}^1, \dots, \mathbf{p}^N).$$

In the last step, we apply the Koksma-Hlawka inequality, where we exploit the properties of variation in the sense of Hardy-Krause for linear functions like $g(x) = 2x - 1$.

This implies that (2.27) holds, whenever we secure

$$1 - \rho \geq 4\mathcal{D}_N^*(\mathbf{p}^1, \dots, \mathbf{p}^N). \quad (2.28)$$

Since $(\mathbf{p}^n)_{n \in \mathbb{N}}$ is a point sequence uniformly distributed modulo one, its star-discrepancy converges to 0 as $N \rightarrow \infty$. Therefore, for any $\rho \in (0, 1)$, there exists a $N \in \mathbb{N}$, such that $\delta > 0$.

Remark 2.10. To better understand the condition on α in Theorem 2.8, we revisit the constant

$$\mathbf{C}_{\text{QMC}}(N, \alpha) = C_{f,K,D}(\mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N))^{1/D} \frac{\alpha^{\frac{D}{\beta}+1}}{C_0(\alpha)}.$$

While this term vanishes as $N \rightarrow \infty$ due to the convergence of the star discrepancy, it grows polynomially in α , with a degree that increases with the dimension D , because $C_0(\alpha)$ converges to a positive constant as $\alpha \rightarrow \infty$.

As a consequence, satisfying the condition

$$(e^{\lambda T} - 1)(\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)) \leq \frac{K(1-\rho)}{2}$$

requires a careful balance between α and N . On the one hand, α must be chosen sufficiently large to ensure that $\mathbf{C}_{\text{Lap}}(\alpha)$ is small. On the other hand, increasing α raises $\mathbf{C}_{\text{QMC}}(N, \alpha)$, so we need to choose a larger number of particles N as compensation. Hence, it is generally not optimal to choose α arbitrarily large. Instead, one should select α as small as possible while keeping $\mathbf{C}_{\text{Lap}}(\alpha)$ sufficiently small, and therefore minimize the required number of particles to control $\mathbf{C}_{\text{QMC}}(N, \alpha)$. Nevertheless, since the star discrepancy converges to zero as $N \rightarrow \infty$, for any fixed choice of λ , T , ρ , and K , there always exists a combination of α and N such that the above condition is satisfied.

3 Proof details for Section 2

In this section, we provide detailed proofs for the auxiliary lemmas from Section 2, which quantify the contributions of the two approximations. We first give the proof of Lemma 2.2. Then, we proof 2.5, which bounds the Quasi-Monte Carlo approximation error for the finite particle system. We then proceed with the proof of Lemma 2.6, which quantifies the Laplace principle error for the finite-concentration system.

Proof of Lemma 2.2. To simplify the structure of $I(g_0)$, we consider the affine change of variables

$$y = z + e^{-\lambda s} \mathfrak{s}(x).$$

Since $\mathfrak{s}$ is a linear mapping from $[0, 1)^D$ onto $[-K, K]^D$, it follows that

$$\mathbf{0}_D < x < \mathbf{1}_D \quad \Leftrightarrow \quad -Ke^{-\lambda s} \mathbf{1}_D < e^{-\lambda s} \mathfrak{s}(x) < Ke^{-\lambda s} \mathbf{1}_D \quad \Leftrightarrow \quad y \in B_{Ke^{-\lambda s}}^\infty(z).$$

In particular, the mapping

$$\Phi : (0, 1)^D \rightarrow B_{Ke^{-\lambda s}}^\infty(z), \quad \Phi(x) := z + e^{-\lambda s} \mathfrak{s}(x).$$

is continuously differentiable on $(0, 1)^D$. Moreover, the Jacobian of Φ is constant and given by

$$D\Phi(x) = e^{-\lambda s} D\mathfrak{s} = 2Ke^{-\lambda s} I_D,$$

where I_D denotes the D -dimensional identity matrix. Consequently, the determinant of the Jacobian is constant and equals

$$\det(D\Phi(x)) = (2K)^D e^{-D\lambda s}.$$

Since $\mathfrak{s}$ is bijective between $[0, 1)^D$ and $[-K, K]^D$, the mapping Φ is a bijective as well, and its inverse is continuous differentiable. Hence, Φ is a diffeomorphism in the sense required for a change of variables. Applying said change of variables yields

$$I(g_0) = \int_{(0, 1)^D} \exp(-\alpha f(z + e^{-\lambda s} \mathfrak{s}(x))) dx = \frac{1}{(2K)^D e^{-D\lambda s}} \int_{B_{Ke^{-\lambda s}}^\infty(z)} \exp(-\alpha f(y)) dy.$$

The first identity holds because the set $[0, 1)^D \setminus (0, 1)^D$ is a set of measure zero and thus does not contribute to the value of the integral.

Since the exponential function is strictly positive, $B_\delta^\infty(x^*) \subseteq B_{Ke^{-\lambda s}}^\infty(z)$ and $e^{-x} \leq 1$ for all $x \geq 0$, we obtain

$$I(g_0) \geq \frac{1}{(2K)^D} \int_{B_\delta^\infty(x^*)} \exp(-\alpha f(y)) dy.$$

Furthermore, we have $\delta \leq \delta_{\text{loc}}$, so assumption (A2) applies, and we can write

$$\int_{B_\delta^\infty(x^*)} \exp(-\alpha f(y)) dy = \exp(-\alpha f(x^*)) \int_{B_\delta^\infty(x^*)} \exp(-\alpha m_f \|y - x^*\|^\beta - \alpha R(y, x^*)) dy$$

To make the asymptotic behavior as $\alpha \rightarrow \infty$ explicit, we perform another change of variables. We introduce the affine transformation

$$w = \alpha^{\frac{1}{\beta}} (y - x^*) \quad \Leftrightarrow \quad y = x^* + \alpha^{-\frac{1}{\beta}} w,$$

under which the integration domain transforms as

$$y \in B_\delta^\infty(x^*) \quad \Leftrightarrow \quad x^* + \alpha^{-\frac{1}{\beta}} w \in B_\delta^\infty(x^*) \quad \Leftrightarrow \quad w \in B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D).$$

Motivated by this rescaling, we define the mapping

$$\Psi : B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D) \rightarrow B_\delta^\infty(x^*), \quad \Psi(w) := x^* + \alpha^{-\frac{1}{\beta}} w.$$

Both $B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)$ and $B_\delta^\infty(x^*)$ are open sets. Since Ψ is affine, it is a diffeomorphism with continuous differentiable inverse and constant Jacobian

$$D\Psi(w) = \alpha^{-\frac{1}{\beta}} I_D,$$

whose determinant is given by

$$\det D\Psi(w) = \alpha^{-\frac{D}{\beta}}.$$

Using the identity

$$\|y - x^*\|^\beta = \left\| x^* + \alpha^{-\frac{1}{\beta}} w - x^* \right\|^\beta = \alpha^{-1} \|w\|^\beta,$$

a change of variables yields

$$\int_{B_\delta^\infty(x^\star)} \exp(-\alpha f(y)) dy = \frac{\alpha^{-\frac{D}{\beta}}}{\exp(\alpha f(x^\star))} C_0(\alpha),$$

where

$$C_0(\alpha) = \int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} \exp\left(-m_f\|w\|^\beta - \alpha R\left(x^\star + \alpha^{-\frac{1}{\beta}}w, x^\star\right)\right) dw.$$

By monotonicity of the exponential function, Assumption (A2), and the fact that $\alpha > 0$, we obtain the upper bound

$$\begin{aligned} \exp\left(-m_f\|w\|^\beta - \alpha R\left(x^\star + \alpha^{-\frac{1}{\beta}}w, x^\star\right)\right) &\leq \exp\left(-m_f\|w\|^\beta + \alpha \left|R\left(x^\star + \alpha^{-\frac{1}{\beta}}w, x^\star\right)\right|\right) \\ &\leq \exp\left(-m_f\|w\|^\beta + \alpha \frac{m_f}{2} \left\|\alpha^{-\frac{1}{\beta}}w\right\|^\beta\right) \\ &= \exp\left(-\frac{m_f}{2}\|w\|^\beta\right). \end{aligned}$$

Analogously, $\exp(-3m_f/2\|w\|^\beta)$ provides a lower bound for this term. Hence, we obtain the bounds

$$\int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} \exp\left(-\frac{3m_f}{2}\|x\|^\beta\right) dx \leq C_0(\alpha) \leq \int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} \exp\left(-\frac{m_f}{2}\|x\|^\beta\right) dx. \quad (3.1)$$

We now study the asymptotic behavior of $C_0(\alpha)$ as $\alpha \rightarrow \infty$ using the dominated convergence theorem. To this end, we rewrite

$$C_0(\alpha) = \int_{\mathbb{R}^D} \underbrace{\exp\left(-m_f\|w\|^\beta - \alpha R\left(x^\star + \alpha^{-\frac{1}{\beta}}w, x^\star\right)\right)}_{=: g_\alpha(w)} \mathbb{1}_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)}(w) dw.$$

By the bounds in (3.1) and the positivity of the exponential function, the integrand g_α is dominated by

$$\exp\left(-\frac{m_f}{2}\|w\|^\beta\right) \mathbb{1}_{\mathbb{R}^D}(w),$$

which is integrable. Indeed, using (2.13) with $b = 0$ and $a = \frac{m_f}{2}$, we obtain

$$\int_{\mathbb{R}^D} \exp\left(-\frac{m_f}{2}\|w\|^\beta\right) dw = \frac{2}{\beta} \pi^{\frac{D}{2}} \left(\frac{m_f}{2}\right)^{-\frac{D}{\beta}} \frac{\Gamma(D/\beta)}{\Gamma(D/2)} < \infty.$$

Moreover, Assumption (A2) implies the upper bound

$$\begin{aligned} \exp\left(-\alpha R\left(x^\star + \alpha^{-\frac{1}{\beta}}w, x^\star\right)\right) &\leq \exp\left(\alpha \left|R\left(x^\star + \alpha^{-\frac{1}{\beta}}w, x^\star\right)\right|\right) \\ &\leq \exp\left(\alpha C_f \left\|\alpha^{-\frac{1}{\beta}}w\right\|^{\beta+\gamma}\right) \\ &= \exp\left(C_f \alpha^{-\frac{\gamma}{\beta}} \|w\|^{\beta+\gamma}\right), \end{aligned}$$

and similarly the lower bound $\exp(-C_f \alpha^{-\gamma/\beta} \|w\|^{\beta+\gamma})$. Since $\beta, \gamma > 0$, both bounds converge point-wise to $\exp(0) = 1$ as $\alpha \rightarrow \infty$. Therefore we have

$$\exp\left(-\alpha R\left(x^\star + \alpha^{-\frac{1}{\beta}}w, x^\star\right)\right) \rightarrow 1.$$

Together with the fact that $B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)$ converges to $\mathbb{R}^D$ as $\alpha \rightarrow \infty$ this yields the point-wise convergence

$$g_\alpha(x) \rightarrow \exp(-m_f\|w\|^\beta) \mathbb{1}_{\mathbb{R}^D}(w).$$

We may therefore apply the dominated convergence theorem to conclude that

$$C_0(\infty) = \lim_{\alpha \rightarrow \infty} \int_{\mathbb{R}^D} g_\alpha(w) dw = \int_{\mathbb{R}^D} \exp(-m_f\|w\|^\beta) dw = \frac{2}{\beta} \pi^{\frac{D}{2}} m_f^{-\frac{D}{\beta}} \frac{\Gamma(D/\beta)}{\Gamma(D/2)}, \quad (3.2)$$

where the final identity follows from (2.13) with $a = m_f$ and $b = 0$.

This yields the desired result

$$I(g_0) \geq \frac{C_0(\alpha)\alpha^{-\frac{D}{\beta}}}{(2K)^D \exp(\alpha f(x^*))}$$

Here, $C_0(\alpha)$ is defined in (2.16), satisfies the bounds in (3.1), and converges to a finite, positive constant according to (3.2). $\square$

We now present the proof of Lemma 2.5.

Proof of Lemma 2.5. Recall the definition of the auxiliary functions g_d in 2.7. Since f is locally Lipschitz continuous and bounded from below, the function g_d is locally Lipschitz on $\mathbb{R}^D$. When restricted to the domain $[-2K, 2K]^D$, it follows that g_d is bounded and globally Lipschitz with some constant $L_{g_d} > 0$.

Moreover, since f is differentiable, the function g_d is differentiable as well. An application of the mean value theorem therefore yields the existence of a point $\zeta \in (-2K, 2K)^D$ such that

$$|g_d(u) - g_d(v)| = \|\nabla g_d(\zeta)\| \|u - v\| \leq \sup_{w \in [-2K, 2K]^D} \|\nabla g_d(w)\| \|u - v\|.$$

By the chain rule, the gradient of g_d is given by

$$\nabla g_d(v) = e_d \exp(-\alpha f(z + ve^{-\lambda s})) - \alpha v_d e^{-\lambda s} \nabla f(z + ve^{-\lambda s}) \exp(-\alpha f(z + ve^{-\lambda s})),$$

where $e_0 = \mathbf{0}_D$ denotes the zero vector and e_d is the d -th canonical basis vector of $\mathbb{R}^D$.

This and $z \in [-K, K]^D$ allows us to bound the Lipschitz constant associated with g_d by

$$L_{g_d} \leq \sup_{w \in [-2K, 2K]^D} \exp(-\alpha f(z + we^{-\lambda s})) \left(\|e_d\| + \alpha |w_d| e^{-\lambda s} \sup_{x \in [-3K, 3K]^D} \|\nabla f(x)\| \right).$$

Since $w \in [-2K, 2K]^D$ and x^* is a global minimizer of f , we obtain the upper bound

$$L_{g_d} \leq \exp(-\alpha f(x^*)) \begin{cases} 1 + 2\alpha K e^{-\lambda s} \|\nabla f\|_\infty, & d = 1, \dots, D, \\ 2\alpha K e^{-\lambda s} \|\nabla f\|_\infty, & d = 0. \end{cases} \quad (3.3)$$

Here, the quantity

$$\|\nabla f\|_\infty = \sup_{x \in [-3K, 3K]^D} \|\nabla f(x)\|$$

denotes the supremum norm of the gradient of f over the relevant domain.

Next, we collect several observations concerning the mapping $\mathfrak{s}$, which will allow us to rewrite $\tilde{F}_{N,\alpha}(z, s)$ in a form that is well suited for a Quasi-Monte Carlo analysis. We begin by noting that the linearity of $\mathfrak{s}$ implies that the components

$$(\mathfrak{s}_d)_\mathcal{V}(u) = (2Ku_d - K)\mathbb{1}_{\{d \in \mathcal{V}\}} + K\mathbb{1}_{\{d \notin \mathcal{V}\}}$$

for $d = 1, \dots, D$ admit particularly simple partial derivatives. Indeed, for $d' = 1, \dots, D$ we have

$$\frac{\partial}{\partial u_{d'}} (\mathfrak{s}_d)_{\{d'\}}(u) = 2K\mathbb{1}_{\{d=d'\}}, \quad \frac{\partial^k}{\partial u_1 \dots \partial u_k} (\mathfrak{s}_d)_\mathcal{V}(u) = 0$$

for all $1 < k \leq D$ and all subsets $\mathcal{V} \subseteq \{1, \dots, D\}$ with $\#\mathcal{V} > 1$. In particular, all mixed and higher-order derivatives vanish. As a consequence, the Hardy-Krause variation of $\mathfrak{s}_d$ reduces to

$$V(\mathfrak{s}_d) = \sum_{\substack{\mathcal{V} \subseteq \{1, \dots, D\} \\ \#\mathcal{V}=1}} V((\mathfrak{s}_d)_\mathcal{V}) = \sum_{d'=1}^D \int_0^1 \left| \frac{\partial}{\partial u_{d'}} (\mathfrak{s}_d)_{\{d'\}}(u) \right| du_{d'} = \int_0^1 2K du_d = 2K.$$

Furthermore, the linearity of $\mathfrak{s}$ also implies that the initial residual satisfies

$$r_0^{m,d} = x_0^{m,d} - \bar{x}_0^{N,d} = \mathfrak{s}_d(\mathbf{p}^m) - \frac{1}{N} \sum_{k=1}^N \mathfrak{s}_d(\mathbf{p}^k) = \mathfrak{s}_d(\mathbf{p}^m) - \mathfrak{s}_d(\bar{\mathbf{p}}^N)$$

for all $m = 1, \dots, N$ and $d = 1, \dots, D$. Substituting this identity into the definition of $\tilde{F}_{N,\alpha}(z, s)$ yields

$$\tilde{F}_{N,\alpha}(z, s) = \frac{1}{N} \sum_{m=1}^N (\mathfrak{s}(\mathbf{p}^m) - \mathfrak{s}(\bar{\mathbf{p}}^N)) \frac{\exp(-\alpha f(z + e^{-\lambda s}[\mathfrak{s}(\mathbf{p}^m) - \mathfrak{s}(\bar{\mathbf{p}}^N)]))}{\frac{1}{N} \sum_{k=1}^N \exp(-\alpha f(z + e^{-\lambda s}[\mathfrak{s}(\mathbf{p}^k) - \mathfrak{s}(\bar{\mathbf{p}}^N)]))}.$$

By construction of the functions g_d , this expression can be written component-wise as

$$(\tilde{F}_{N,\alpha}(z, s))_d = \frac{\frac{1}{N} \sum_{m=1}^N g_d(\mathfrak{s}(\mathbf{p}^m) - \mathfrak{s}(\bar{\mathbf{p}}^N))}{\frac{1}{N} \sum_{k=1}^N g_0(\mathfrak{s}(\mathbf{p}^k) - \mathfrak{s}(\bar{\mathbf{p}}^N))}, \quad (\tilde{F}_{\infty,\alpha}(z, s))_d := \frac{\int_{[0,1]^D} g_d(\mathfrak{s}(x)) dx}{\int_{[0,1]^D} g_0(\mathfrak{s}(x)) dx}.$$

for $d = 1, \dots, D$.

We now recall the definition of $I(g_d)$ in (2.8) and introduce, for $d = 0, \dots, D$,

$$I_N(g_d) = \frac{1}{N} \sum_{m=1}^N g_d(\mathfrak{s}(\mathbf{p}^m) - \mathfrak{s}(\bar{\mathbf{p}}^N)), \quad I'_N(g_d) = \frac{1}{N} \sum_{m=1}^N g_d(\mathfrak{s}(\mathbf{p}^m)).$$

The absolute difference $|I(g_d) - I'_N(g_d)|$ can be estimated by the Koksma-Hlawka-type inequality. More precisely, applying Theorem 2.4 to the integrand $g_d \circ \mathfrak{s} : [0, 1]^D \rightarrow \mathbb{R}$ yields

$$|I(g_d) - I'_N(g_d)| \leq 4\omega(g_d \circ \mathfrak{s}; \mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N)^{1/D}),$$

where $\mathcal{D}_N^* = \mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N)$ denotes the star discrepancy of the point set $\{\mathbf{p}_1, \dots, \mathbf{p}_N\}$. Note that the composition $g_d \circ \mathfrak{s}$ is continuous, since g_d is continuous and $\mathfrak{s}$ is linear. Moreover, the global Lipschitz continuity of g_d and the linearity of $\mathfrak{s}$ imply

$$|g_d(\mathfrak{s}(u)) - g_d(\mathfrak{s}(v))| \leq L_{g_d} \|\mathfrak{s}(u) - \mathfrak{s}(v)\| = 2KL_{g_d} \|u - v\|.$$

Since $\|x\| \leq \sqrt{D}\|x\|_\infty$ holds for any $x \in \mathbb{R}^D$, the modulus of continuity of $g_d \circ \mathfrak{s}$ can be bounded by

$$\omega(g_d \circ \mathfrak{s}; (\mathcal{D}_N^*)^{1/D}) = \sup_{\substack{u, v \in [0,1]^D \\ \|u-v\|_\infty \leq (\mathcal{D}_N^*)^{1/D}}} |g_d(\mathfrak{s}(u)) - g_d(\mathfrak{s}(v))| \leq 2\sqrt{D}KL_{g_d}(\mathcal{D}_N^*)^{1/D}.$$

Altogether, this yields the estimate

$$|I'_N(g_d) - I_N(g_d)| \leq 8\sqrt{D}KL_{g_d}(\mathcal{D}_N^*)^{1/D}. \quad (3.4)$$

Exploiting again the linearity of $\mathfrak{s}$ and the fact that $\mathbf{p}^m$, and hence also $\bar{\mathbf{p}}^N$, take values in the unit cube $[0, 1]^D$, we observe that

$$|\mathfrak{s}_d(\mathbf{p}^m) - \mathfrak{s}_d(\bar{\mathbf{p}}^N)| = 2K |\mathbf{p}^{m,d} - \bar{\mathbf{p}}^{N,d}| \leq 2K.$$

Therefor, we immediately obtain an estimate for the absolute ratio of $I_N(g_d)$ and $I_N(g_0)$, namely

$$\left| \frac{I_N(g_d)}{I_N(g_0)} \right| \leq \frac{\frac{1}{N} \sum_{m=1}^N |\mathfrak{s}_d(\mathbf{p}^m) - \mathfrak{s}_d(\bar{\mathbf{p}}^N)| \exp(-\alpha f(z + e^{-\lambda t}[\mathfrak{s}(\mathbf{p}^m) - \mathfrak{s}(\bar{\mathbf{p}}^N)]))}{\frac{1}{N} \sum_{m=1}^N \exp(-\alpha f(z + e^{-\lambda t}[\mathfrak{s}(\mathbf{p}^m) - \mathfrak{s}(\bar{\mathbf{p}}^N)]))} \leq 2K. \quad (3.5)$$

Next, we estimate the absolute difference $|I'_N(g_d) - I_N(g_d)|$ by exploiting the Lipschitz continuity of L_{g_d} . Indeed, we compute

$$|I'_N(g_d) - I_N(g_d)| = \left| \frac{1}{N} \sum_{m=1}^N g_d(\mathfrak{s}(\mathbf{p}^m)) - g_d(\mathfrak{s}(\mathbf{p}^m) - \mathfrak{s}(\bar{\mathbf{p}}^N)) \right| \leq \frac{1}{N} \sum_{m=1}^N L_{g_d} \|\mathfrak{s}(\bar{\mathbf{p}}^N)\|$$

To bound the remaining term, we first observe that

$$\int_{[0,1]^D} \mathfrak{s}_d(x) dx = \int_{[0,1]^{D-1}} \int_0^1 2Kx_d - K dx_d dx_{-d} = \int_{[0,1]^{D-1}} 0 dx_{-d} = 0,$$

where x_{-d} denotes all components of the vector x except for the d -th one. Since $\mathfrak{s}_d$ is linear and hence continuous, and since its variation in the sense of Hardy-Krause satisfies $V(\mathfrak{s}_d) = 2K$, we may apply the Koksma-Hlawka inequality from Theorem 2.4 to obtain

$$\|\mathfrak{s}(\bar{\mathbf{p}}^N)\| = \sqrt{\sum_{d=1}^D \left| \frac{1}{N} \sum_{m=1}^N \mathfrak{s}_d(\mathbf{p}^m) - \int_{[0,1]^D} \mathfrak{s}_d(x) dx \right|^2} \leq \sqrt{\sum_{d=1}^D |2K\mathcal{D}_N^*|^2} = 2\sqrt{D}K\mathcal{D}_N^*.$$

Combining the above estimates yields

$$|I(g_d) - I'_N(g_d)| \leq 2\sqrt{D}KL_{g_d}\mathcal{D}_N^*. \quad (3.6)$$

With the construction of the quantities $I_N(g_d)$ and $I(g_d)$, we can bound the desired component-wise absolute difference between $\tilde{F}_{N,\alpha}(z, s)$ and $\tilde{F}_{\infty,\alpha}(z, s)$. More precisely, for a fixed component index $d \in \{1, \dots, D\}$, we obtain

$$\begin{aligned} \left| (\tilde{F}_{N,\alpha}(z, s) - \tilde{F}_{\infty,\alpha}(z, s))_d \right| &= \left| \frac{I_N(g_d)}{I_N(g_0)} - \frac{I(g_d)}{I(g_0)} \right| \\ &\leq \left| \frac{I_N(g_d)}{I(g_0)} - \frac{I(g_d)}{I(g_0)} \right| + \left| I_N(g_d) \left(\frac{1}{I_N(g_0)} - \frac{1}{I(g_0)} \right) \right| \\ &= \frac{|I_N(g_d) - I(g_d)|}{I(g_0)} + \left| \frac{I_N(g_d)}{I_N(g_0)} \right| \frac{|I(g_0) - I_N(g_0)|}{I(g_0)} \\ &\leq \frac{1}{I(g_0)} |I(g_d) - I_N(g_d)| + \frac{2K}{I(g_0)} |I(g_0) - I_N(g_0)| \end{aligned}$$

where in the last step we used the bound (3.5) on the ratio $|I_N(g_d)/I_N(g_0)|$.

Next, for $d = 0, \dots, D$, we combine the estimates obtained in (3.6) and (3.4) to control the difference between $I(g_d)$ and $I_N(g_d)$. Indeed, we have

$$\begin{aligned} |I(g_d) - I_N(g_d)| &\leq |I(g_d) - I'_N(g_d)| + |I'_N(g_d) - I_N(g_d)| \\ &\leq 2\sqrt{D}KL_{g_d} \left(\mathcal{D}_N^* + 4(\mathcal{D}_N^*)^{1/D} \right) \\ &\leq 10\sqrt{D}KL_{g_d}(\mathcal{D}_N^*)^{1/D}. \end{aligned}$$

since $\mathcal{D}_N^* < 1$. Substituting this estimate into the previous bound yields

$$\begin{aligned} \left| (\tilde{F}_{N,\alpha}(z, s) - \tilde{F}_{\infty,\alpha}(z, s))_d \right| &\leq \frac{10\sqrt{D}K(\mathcal{D}_N^*)^{1/D}}{I(g_0)} (L_{g_d} + 2KL_{g_0}) \\ &\leq \frac{10\sqrt{D}K(\mathcal{D}_N^*)^{1/D} \alpha^{\frac{D}{\beta}}}{C_0(\alpha) \exp(-\alpha f(x^*))} (2K)^D (L_{g_d} + 2KL_{g_0}), \end{aligned}$$

where the last step follows from assumptions 2.1, the inclusion $B_\delta^\infty(x^*) \subseteq B_{Ke^{-\lambda s}}^\infty(z)$ and an application of Lemma 2.2. Here, $C_0(\alpha)$ is defined in (2.16), satisfies the bounds in (3.1), and converges to the finite constant given by (2.17).

We now insert the bounds for the Lipschitz constants L_{g_d} and L_{g_0} given in (3.3) and use $e^{-x} \leq 1$ for all $x \geq 1$ as well as $\alpha \geq 1$ to obtain

$$(L_{g_d} + 2KL_{g_0}) \leq \exp(-\alpha f(x^*)) \alpha (1 + 2K\|\nabla f\|_\infty (1 + 2K)).$$

Altogether, we therefore arrive at the estimate

$$\begin{aligned} \left| (\tilde{F}_{N,\alpha}(z, s) - \tilde{F}_{\infty,\alpha}(z, s))_d \right| &\leq 10\sqrt{D}(2K)^{D+1} (1 + 2K\|\nabla f\|_\infty (1 + 2K)) (\mathcal{D}_N^*)^{1/D} \frac{\alpha^{\frac{D}{\beta}+1}}{C_0(\alpha)} \\ &= C_{f,K,D} (\mathcal{D}_N^*)^{1/D} \frac{\alpha^{\frac{D}{\beta}+1}}{C_0(\alpha)}, \end{aligned}$$

where

$$C_{f,K,D} := 10\sqrt{D}(2K)^{D+1} (1 + 2K\|\nabla f\|_\infty (1 + 2K)) \quad (3.7)$$

only depends on f , K and D . Recall that the sequence $\mathcal{S} = (\mathbf{p}^n)_{n \in \mathbb{N}}$ is uniformly distributed modulo one if and only if its star discrepancy $\mathcal{D}_N^*(\mathbf{p}^1, \dots, \mathbf{p}^N)$ converges to zero as $N \rightarrow \infty$. Consequently, for each component $d \in \{1, \dots, D\}$,

$$\lim_{N \rightarrow \infty} \left| (\tilde{F}_{N,\alpha}(z, s) - \tilde{F}_{\infty,\alpha}(z, s))_d \right| = 0,$$

which implies the convergence of the full vector-valued map

$$\lim_{N \rightarrow \infty} \tilde{F}_{N,\alpha}(z, s) = \tilde{F}_{\infty,\alpha}(z, s).$$

□

We now present the proof of Lemma 2.6.

Proof of Lemma 2.6. By (2.20), and using that the integral of the constant function 1 over $[0, 1)^D$ equals 1, we already know that the d -th component of $F_{\infty,\alpha}(z, s)$ can be written as

$$(F_{\infty,\alpha}(z, s))_d = \frac{\int_{[0,1)^D} (z_d + e^{-\lambda s} \mathbf{s}_d(x) - x_d^*) \exp(-\alpha f(z + e^{-\lambda s} \mathbf{s}(x))) dx}{\int_{[0,1)^D} \exp(-\alpha f(z + e^{-\lambda s} \mathbf{s}(x))) dx}.$$

Following the arguments in the proof of Lemma 2.2, a change of variables Φ in the denominator leads to the representation

$$\int_{[0,1)^D} \exp(-\alpha f(z + e^{-\lambda s} \mathbf{s}(x))) dx = \frac{1}{(2K)^D e^{-D\lambda s}} \int_{B_{Ke^{-\lambda s}}^\infty(z)} \exp(-\alpha f(y)) dy.$$

Similarly, applying Φ to the numerator gives

$$\int_{[0,1)^D} \frac{z_d + e^{-\lambda s} \mathbf{s}_d(x) - x_d^*}{\exp(\alpha f(z + e^{-\lambda s} \mathbf{s}(x)))} dx = \frac{1}{(2K)^D e^{-D\lambda s}} \int_{B_{Ke^{-\lambda s}}^\infty(z)} (y_d - x_d^*) \exp(-\alpha f(y)) dy$$

As a result, the expression for $(F_{\infty,\alpha}(z, s))_d$ simplifies to

$$(F_{\infty,\alpha}(z, s))_d = \frac{\int_{B_{Ke^{-\lambda s}}^\infty(z)} (y_d - x_d^*) \exp(-\alpha f(y)) dy}{\int_{B_{Ke^{-\lambda s}}^\infty(z)} \exp(-\alpha f(y)) dy}. \quad (3.8)$$

The denominator of (3.8) was likewise analyzed in the proof of Lemma 2.2. By employing a further change of variables Ψ , we obtain the lower bound

$$\int_{B_{Ke^{-\lambda s}}^\infty(z)} \exp(-\alpha f(y)) dy \geq \frac{C_0(\alpha) \alpha^{-\frac{D}{\beta}}}{\exp(\alpha f(x^*))},$$

where $C_0(\alpha)$ is defined in (2.16), satisfies the bounds in (3.1), and converges to the positive constant given in (3.2).

We now determine analogous results for the numerator of (3.8). By assumption, there exists a $\delta \leq \delta_{\text{loc}}$ such that $B_\delta^\infty(x^*) \subseteq B_{Ke^{-\lambda s}}^\infty(z)$. Fixing such a δ , we define

$$D_R(z, s) := B_{Ke^{-\lambda s}}^\infty(z) \setminus B_\delta^\infty(x^*).$$

We now split the domain of integration in (3.8) into the disjoint subsets $B_\delta^\infty(x^*)$ and $D_R(z, s)$, and investigate the resulting integrals separately. This decomposition separates the contribution coming from a neighborhood of the minimizer x^* from the contribution away from the minimizer.

We first consider the integral over the neighborhood of the minimizer. Using Assumption (A2) and applying the change of variables induced by the mapping Ψ , we obtain

$$\int_{B_\delta^\infty(x^*)} (y_d - x_d^*) \exp(-\alpha f(y)) dy = \frac{\alpha^{-\frac{D+1}{\beta}}}{\exp(\alpha f(x^*))} C_1(\alpha), \quad (3.9)$$

where

$$C_1(\alpha) := \int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} w_d \exp\left(-m_f \|w\|^\beta - \alpha R\left(x^* + \alpha^{-\frac{1}{\beta}} w, x^*\right)\right) dw. \quad (3.10)$$

As in the analysis of $C_0(\alpha)$ in Lemma 2.2, we derive bounds on $C_1(\alpha)$ and determine its limit as $\alpha \rightarrow \infty$. We begin by estimating $|C_1(\alpha)|$. By the mean value theorem, for any $v \in \mathbb{R}$ there exists ζ between 0 and v , such that for the continuous function $h(u) = \exp(-u)$,

$$|\exp(-v) - 1| = |\exp(-v) - \exp(-0)| = \exp(-\zeta)|v|.$$

Since the exponential function is monotonically increasing and ζ lies between 0 and v , this yields

$$|\exp(-v) - 1| \leq \exp(|v|)|v|.$$

Next, we exploit the symmetry of the integration domain of $C_1(\alpha)$ and of the function $\exp(-m_f \|w\|^\beta)$. Each component w_d ranges over the symmetric interval $[-\delta\alpha^{1/\beta}, \delta\alpha^{1/\beta}]$. Writing $w = (w_d, w_{-d})$ and applying Fubini, we obtain

$$\int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} w_d \exp(-m_f \|w\|^\beta) dw = \int_{-\delta\alpha^{1/\beta}}^{\delta\alpha^{1/\beta}} \dots \int_{-\delta\alpha^{1/\beta}}^{\delta\alpha^{1/\beta}} w_d \exp(-m_f \|w\|^\beta) dw_d dw_{-d} = 0.$$

In the last step, we use that for fixed w_{-d} , the factor $\exp(-m_f \|w\|^\beta)$ depends on w_d only through $|w_d|$. Hence, the contributions corresponding to w_d and $-w_d$ cancel each other. As a result, the inner integral vanishes and the entire integral is equal to zero.

Using this identity and the inequality above, we may write

$$\begin{aligned} |C_1(\alpha)| &= \left| \int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} w_d \exp(-m_f \|w\|^\beta) \left(\exp\left(-\alpha R\left(x^* + \alpha^{-\frac{1}{\beta}} w; x^*\right)\right) - 1 \right) dw \right| \\ &\leq \int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} |w_d| \exp(-m_f \|w\|^\beta) \exp\left(\alpha \left| R\left(x^* + \alpha^{-\frac{1}{\beta}} w; x^*\right) \right| \right) \alpha \left| R\left(x^* + \alpha^{-\frac{1}{\beta}} w; x^*\right) \right| dw \end{aligned}$$

Using assumption (A2), the fact that all terms are nonnegative, and estimating $|w_d| \leq \|w\|$, we further compute

$$\begin{aligned} |C_1(\alpha)| &\leq \int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} |w_d| \exp(-m_f \|w\|^\beta) \exp\left(\frac{m_f}{2} \|w\|^\beta\right) \alpha^{-\frac{\gamma}{\beta}} C_f \|w\|^{\beta+\gamma} dw \\ &\leq C_f \alpha^{-\frac{\gamma}{\beta}} \int_{\mathbb{R}^D} \|w\|^{1+\beta+\gamma} \exp\left(-\frac{m_f}{2} \|w\|^\beta\right) dw. \end{aligned}$$

Applying (2.13) with $a = \frac{m_f}{2}$ and $b = 1 + \beta + \gamma$, yields the final bound

$$|C_1(\alpha)| \leq C_f \alpha^{-\frac{\gamma}{\beta}} \frac{2}{\beta} \pi^{\frac{D}{2}} \left(\frac{m_f}{2}\right)^{-\frac{D+1+\beta+\gamma}{\beta}} \frac{\Gamma((D+1+\beta+\gamma)/\beta)}{\Gamma(D/2)}. \quad (3.11)$$

Since $\beta, \gamma > 0$, this final bound converges to 0 as $\alpha \rightarrow \infty$ and therefore

$$C_1(\infty) = 0. \quad (3.12)$$

It remains to estimate the contribution of the numerator over the remainder set $D_R(z, s)$. Since $|y_d - x_d^*|$ is non-negative and integrable on the compact, path-connected set $\overline{B_{Ke-\lambda s}^\infty}(z) \setminus B_\delta^\infty(x^*)$ and $\exp(-\alpha f(y))$ is continuous, the mean value theorem for integrals implies that there exists a $\zeta \in D_R(z, s)$ such that

$$\begin{aligned} \left| \int_{D_R(z, s)} (y_d - x_d^*) \exp(-\alpha f(y)) dy \right| &\leq \int_{\overline{B_{Ke-\lambda s}^\infty}(z) \setminus B_\delta^\infty(x^*)} |y_d - x_d^*| \exp(-\alpha f(y)) dy \\ &= \exp(-\alpha f(\zeta)) \int_{D_R(z, s)} |y_d - x_d^*| dy. \end{aligned}$$

Since ζ lies outside the open δ -neighborhood of the minimizer x^* , assumption (A1) yields

$$\left| \int_{D_R(z,t)} (y_d - x_d^*) \exp(-\alpha f(y)) dy \right| \leq \exp(-\alpha f(x^*)) \exp(-\alpha c) \int_{\overline{B}_{Ke^{-\lambda s}}^\infty(z)} |y_d - x_d^*| dy,$$

where in the last step we enlarged the domain of integration, which is admissible since the integrand is non-negative.

Since $x^* \in B_\delta^\infty(x^*) \subseteq \overline{B}_{Ke^{-\lambda s}}^\infty(z)$, we have

$$|y_d - x_d^*| \leq 2Ke^{-\lambda s}$$

for all $y \in \overline{B}_{Ke^{-\lambda s}}^\infty(z)$, and using that the volume of the cube is $(2Ke^{-\lambda s})^D$, we conclude

$$\left| \int_{D_R(z,t)} (y_d - x_d^*) \exp(-\alpha f(y)) dy \right| \leq \exp(-\alpha f(x^*)) \exp(-\alpha c) (2K)^{D+1} e^{-\lambda s(D+1)}. \quad (3.13)$$

We now have all ingredients in place to conclude the proof by combining the previous estimates. Specifically, by applying the lower bound for the denominator from Lemma 2.2, the estimates (3.9) for the $B_\delta^\infty(x^*)$ -contribution and (3.13) for the $D_R(z, s)$ -contribution to the numerator, we obtain

$$|(F_{\infty,\alpha}(z, s))_d| \leq \frac{\alpha^{-\frac{1}{\beta}} |C_1(\alpha)|}{C_0(\alpha)} + \frac{\exp(-\alpha c)}{C_0(\alpha) \alpha^{-\frac{D}{\beta}}} (2K)^{D+1} e^{-\lambda s(D+1)}.$$

Using the bound (3.11) on $|C_1(\alpha)|$, we finally arrive at

$$|(F_{\infty,\alpha}(z, s))_d| \leq \frac{\alpha^{-\frac{1+\gamma}{\beta}}}{C_0(\alpha)} \tilde{C}_{f,D,\beta,\gamma} + \frac{\alpha^{\frac{D}{\beta}}}{C_0(\alpha) \exp(\alpha c)} (2K)^{D+1} e^{-\lambda s(D+1)},$$

where

$$\tilde{C}_{f,D,\beta,\gamma} := C_f \frac{2}{\beta} \pi^{\frac{D}{2}} \left(\frac{m_f}{2} \right)^{-\frac{D+1+\beta+\gamma}{\beta}} \frac{\Gamma((D+1+\beta+\gamma)/\beta)}{\Gamma(D/2)}.$$

By (3.2), the quantity $C_0(\alpha)$ converges to a finite constant as $\alpha \rightarrow \infty$. Moreover, since the exponential term $\exp(\alpha c)$ grows faster than any polynomial in α , both terms on the right-hand side converge to zero. Combining all components, we conclude that

$$\lim_{\alpha \rightarrow \infty} F_{\infty,\alpha}(z, s) = \mathbf{0}_D.$$

□

4 Numerical Experiments for the QMC²BO Method

In this section, we investigate the error bound established in Theorem 2.8 numerically by understanding how the deviation $\|\bar{x}_t^N - x^*\|$ behaves and how it compares to the exponentially decaying term $e^{-\lambda t} \|\bar{x}_0^N - x^*\|$.

We consider five benchmark functions with qualitatively different properties to investigate the behavior of the QMC²BO method under various optimization challenges. For this purpose, the *Rastrigin*, *Rosenbrock*, and *Styblinski-Tang functions* [27, Optimization Test Problems] are defined as

$$\begin{aligned} f_{\text{Ra}}(x) &= \sum_{d=1}^D 10 + x_d^2 - 10 \cos(2\pi x_d), & f_{\text{Ro}}(x) &= \sum_{d=1}^{D-1} (1 - x_d)^2 + 100(x_{d+1} - x_d^2)^2, \\ f_{\text{ST}}(x) &= \frac{1}{2} \sum_{d=1}^D x_d^4 - 16x_d^2 + 5x_d. \end{aligned} \quad (4.1)$$

All three functions have a unique global minimizer, differing only in its location relative to the origin. The Rastrigin function attains its minimizer at $x^* = \mathbf{0}_D$, the Rosenbrock function at $x^* = \mathbf{1}_D$, and the Styblinski-Tang function at $x^* \approx -2.903534 \mathbf{1}_D$. In addition, these functions differ in their number of

local minima. The Rastrigin function has many regularly distributed local minima. The Styblinski-Tang function possesses three additional local minima, while the Rosenbrock function has only a single global minimum. However, it is flat and narrow around it.

The *Zhou function* [27, Emulation/Prediction Test Problems] and the *fractional power function* are given by

$$f_{\text{Zhou}}(x) = -\frac{10^D}{2} \left(\phi \left(10 \left(x - \frac{1}{3} \right) \right) + \phi \left(10 \left(x - \frac{2}{3} \right) \right) \right), \quad f_{\text{FP}}(x) = \sum_{d=1}^D |x_d|^{\frac{1}{2}}, \quad (4.2)$$

where $\phi(x) = (2\pi)^{-D/2} \exp(-0.5\|x\|^2)$. The Zhou function has two global minimizers at $\frac{1}{3}\mathbf{1}_D$ and $\frac{2}{3}\mathbf{1}_D$, while the fractional power function has its global minimizer at $x^* = \mathbf{0}_D$ but is not differentiable at x^* . Both functions therefore violate assumptions of Theorem 2.8 and test the robustness of the QMC²BO method beyond the theoretical setting in Theorem 2.8.

Unless stated otherwise, all experiments are conducted as follows. The dimension is $D = 2$. The drift parameter is set to $\lambda = 1$ and the concentration parameter to $\alpha = 100$. The system is initialized with $N = 2^7$ particles, generated by sampling Quasi-Monte Carlo (Sobol [26]) points in the cube $[-K, K]^D$ with $K = 4$ using the `scipy.stats.qmc` module from SciPy [31]. For the time evolution, we employ an Euler discretization scheme [18, p. 277] with step size $\Delta t = 0.05$, resulting in a total time horizon of $T = 4.95$, corresponding to 100 time points (epochs). The full implementation, together with additional experiments, is available in the accompanying Git repository [cite](#).

To evaluate the bound from Theorem 2.8, we compare in Figure 1, for each of the five benchmark functions, the quantity $\|\bar{x}_t^N - x^*\|$ (solid lines) with the term $e^{-\lambda t} \|\bar{x}_0^N - x^*\|$ (dashed lines). Each dashed line represents the unavoidable contribution of the initial sampling, which decays exponentially but cannot be reduced by improving any approximation in Section 2, and can only be decreased through a better initialization that requires prior knowledge of the minimizer x^* .

By comparing both quantities, we quantify how much of the observed error is explained purely by the contribution of the initial sampling and how much is due to the additional term in the bound from Theorem 2.8 involving $\mathcal{C}_{\text{QMC}}(N, \alpha)$ and $\mathcal{C}_{\text{Lap}}(\alpha)$, which captures the effect of the Quasi-Monte Carlo approximation (finite N) and the Laplace principle (finite α). Importantly, we do not explicitly evaluate these constants, but instead analyze their practical impact across all benchmark functions.

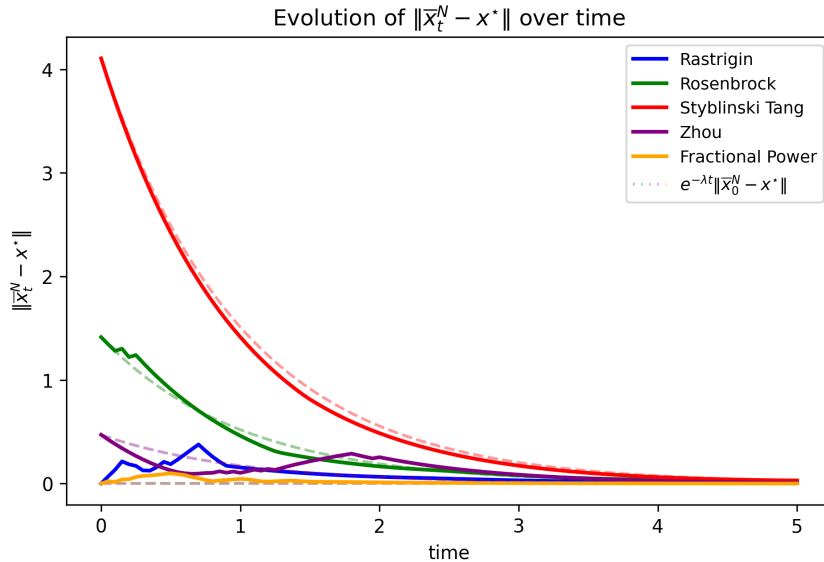


Figure 1: Evolution of $\|\bar{x}_t^N - x^*\|$ (solid lines) and the term $e^{-\lambda t} \|\bar{x}_0^N - x^*\|$ from Theorem 2.8 (dashed lines) for the QMC²BO method 2.7 applied to all benchmark functions.

The main observation from Figure 1 is that, for all functions considered, the average particle position converges to the respective minimizer within the observed time horizon.

Comparing the Rastrigin, Rosenbrock and Styblinski-Tang functions, whose minimizers are located at increasing distance from the origin, we observe that the initial deviation $\|\bar{x}_0^N - x^*\|$ grows accordingly. This is a direct consequence of the Quasi-Monte Carlo initialization in the cube $[-K, K]^D$, whose average

is close to $\mathbf{0}_D$. At the same time, a larger initial distance leads to a closer alignment of $\|\bar{x}_t^N - x^*\|$ with $e^{-\lambda t} \|\bar{x}_0^N - x^*\|$. In particular, for the Styblinski-Tang function, the deviation of the average particle to the minimizer is almost entirely explained by the contribution of the initial sampling, indicating a small effect of finite N and α . A similar behavior is observed for the Rosenbrock function, although there is a short phase in the beginning, in which the effects of finite N and finite α are visible.

In contrast, for the Rastrigin function, whose minimizer is located at the origin, the contribution of the initial particle configuration is close to 0. As a result, $\|\bar{x}_t^N - x^*\|$ is dominated by the additional term in the bound from Theorem 2.8 involving $\mathbf{C}_{\text{QMC}}(N, \alpha)$ and $\mathbf{C}_{\text{Lap}}(\alpha)$, indicating a large effect of finite N and finite α . This effect is especially visible for the first 20 time steps. Afterwards, $\|\bar{x}_t^N - x^*\|$ decreases exponentially. A similar but less pronounced effect is observed for the fractional power function, which also has its minimizer at the origin. This effect might be smaller due to the absence of local minima, which complicate the convergence to the minimizer for the Rastrigin function.

Finally, for the Zhou function with two global minimizers, convergence towards one of them is observed. However, the presence of multiple global minimizers can lead to a temporary increases of $\|\bar{x}_t^N - x^*\|$, even after an initial decay. This behavior can be interpreted as an additional coordination phase, during which the particles agree on one of the two minimizers. This indicates that even when assumptions of Theorem 2.8, such as uniqueness of the minimizer and differentiability, are violated, convergence towards a minimizer can still be observed in practice.

To further investigate the influence of the Quasi-Monte Carlo approximation and the Laplace principle, we focus on the Rastrigin function. In Figure 2, we consider the same setup while varying the number of particles $N \in \{2^5, 2^6, 2^7, 2^8\}$ (left) and the concentration parameter α (right).

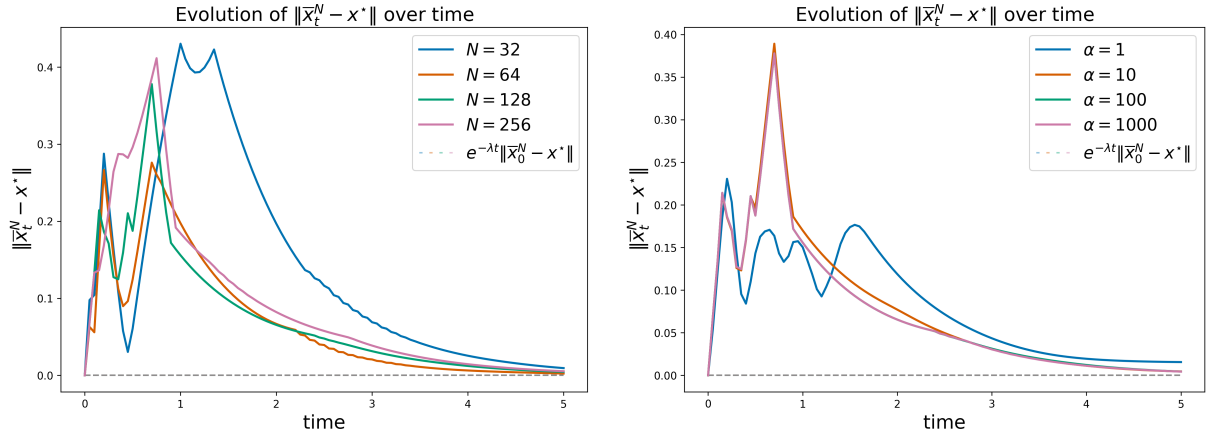


Figure 2: Evolution of $\|\bar{x}_t^N - x^*\|$ (solid lines) and the term $e^{-\lambda t} \|\bar{x}_0^N - x^*\|$ from Theorem 2.8 (dashed lines) for the QMC²BO method 2.7 applied to the Rastrigin function, for varying N (left) and α (right).

Figure 2 shows that the deviation $\|\bar{x}_t^N - x^*\|$ does not decrease monotonically in N . While for $N = 32$ the deviation is significantly larger, intermediate values such as $N = 64$ can have smaller deviations than scenarios with bigger N . This suggests that increasing N does not necessarily improve convergence. This might be caused by an increased difficulty in coordinating larger particle systems.

Across all choices of N , two phases can be identified. In an initial *coordination phase* ($t \in [0, t']$ for some $t' \in (0, T)$), $\|\bar{x}_t^N - x^*\|$ consists mainly of the finite N and α effects, leading to an initial increase in the deviation. Once this phase is overcome, all trajectories decay exponentially towards the minimizer. In this second phase, $\|\bar{x}_t^N - x^*\|$ effectively consists of $e^{-\lambda t} \|\bar{x}_{t'}^N - x^*\|$ for $t \in (t', T]$, indicating that the long-term behavior becomes largely independent of the specific number of particles N .

These two-phases are also observed when varying α . However, in contrast to N , the trajectories are much less sensitive to its value. Apart from very small choices such as $\alpha = 1$, all curves behave nearly identically. This indicates that the impact of N on the dynamics is more influential and moderate values of α already suffice.

In a next step, we investigate the influence of the parameter K on the convergence towards the minimizer. To this end, we consider the Styblinski-Tang function, whose global minimizer, which is located at $x^* \approx -2.903534 \mathbf{1}_D$, lies relatively close to the boundary of $[-K, K]^D$ for $K = 4$. In Figure 3, we examine how well the QMC²BO method approaches the minimizer for $K \in \{2, 2.9, 3, 4\}$. For the two smallest values of K , the Quasi-Monte Carlo points are generated in a region that does not cover x^* .

We observe a clear trend: in all cases, the average particle converges exponentially fast to a consensus

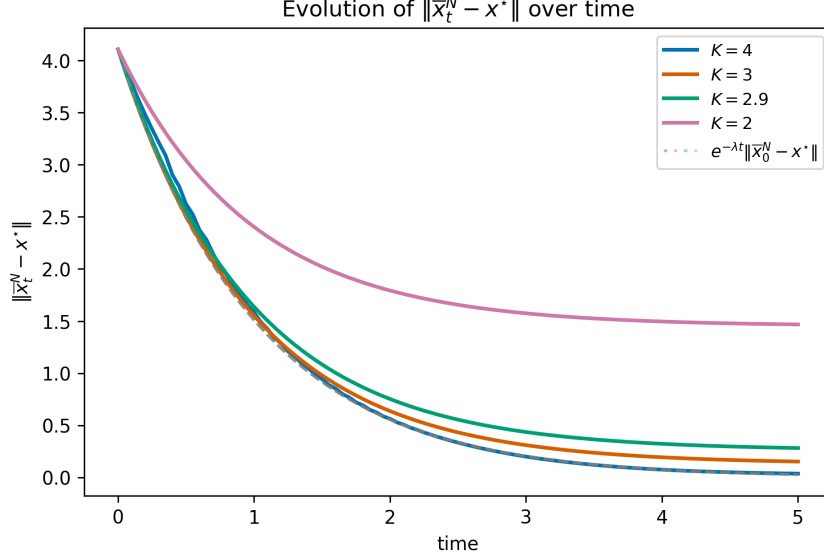


Figure 3: Evolution of $\|\bar{x}_t^N - x^*\|$ (solid lines) and the term $e^{-\lambda t} \|\bar{x}_0^N - x^*\|$ from Theorem 2.8 (dashed lines) for the QMC²BO method 2.7 applied to the Styblinski-Tang function, for varying K .

point, which gets closer to x^* as K increases, i.e., as the Quasi-Monte Carlo initialization covers a larger region. For $K = 2$ and $K = 2.9$, convergence to the minimizer is impossible, since in a purely deterministic system the particles remain within the convex hull of their initial positions. Consequently, the best achievable approximation of x^* is given by the point in this convex hull closest to x^* . Indeed, QMC²BO converges near this point.

This limitation motivates the introduction of stochasticity. In contrast to the deterministic setting, the stochastic CBO dynamics enables particles to leave the convex hull, thereby exploring regions of the landscape that are not covered by the initial configuration. This allows the system to find the minimizer even if it is outside the initial sampling domain.

To investigate stochastic effects, we repeat the previous experiment using the stochastic CBO system introduced in the introduction, i.e., the consensus dynamics with interaction weights (1.2) and an additional stochastic perturbation (1.3), where $\mathcal{S}$ takes the anisotropic form. The system is discretized using the Euler-Maruyama scheme [18, p. 340] with step size $\Delta t = 0.05$. In Figure 4, we show the result of a single run for different values of K . To investigate the influence of randomness, we compare two noise levels, a low value $\sigma = 1$ (left) and a higher value $\sigma = 1.75$ (right).

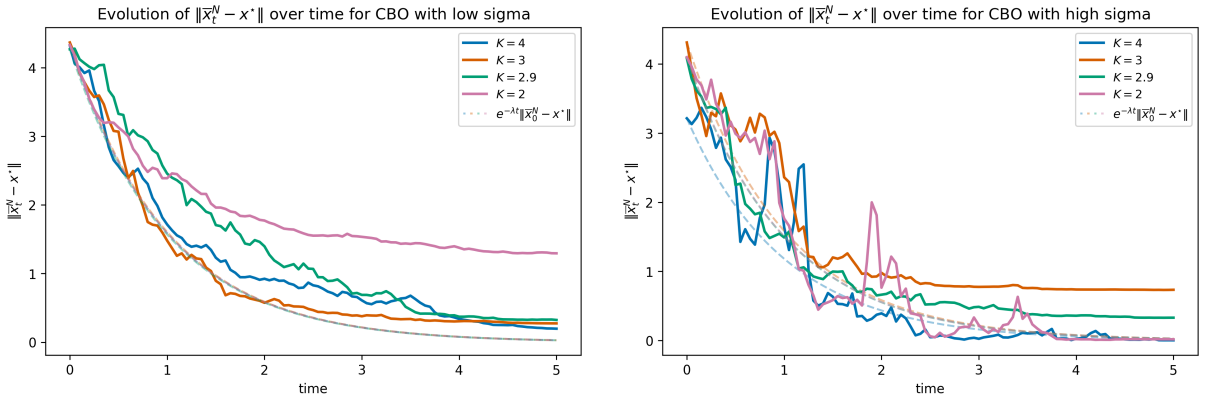


Figure 4: Evolution of $\|\bar{x}_t^N - x^*\|$ (solid lines) and the term $e^{-\lambda t} \|\bar{x}_0^N - x^*\|$ from Theorem 2.8 (dashed lines) for the CBO method (1.2)-(1.3) applied to the Rastrigin function, for varying K and $\sigma = 1$ (left) and $\sigma = 1.75$ (right).

For the case with low σ , we observe a behavior similar to QMC²BO. As K decreases, the particle average becomes a worse approximation of the minimizer. However, the deviation $\|\bar{x}_t^N - x^*\|$ now decays

with small fluctuations over time due to stochastic perturbation.

In contrast, for the case with high σ , the fluctuations are significantly stronger and the quality of the average particle as an approximation of the minimizer varies with respect to K . While $K = 4$ and, notably, $K = 2$ yield accurate approximations, even though $K = 2$ corresponds to the smallest sampling region, the cases $K = 2.9$ and $K = 3$ perform worse than in the deterministic QMC²BO setting. These observations indicate that stochasticity can help overcome the drawbacks of a small initial sampling domain, but may also worsen the approximation when the sampling domain is already sufficiently large for QMC²BO.

We emphasize that these findings are based on single realizations of the CBO dynamics and may vary across runs. To obtain a more robust comparison between QMC²BO and stochastic CBO with different noise levels, we evaluate all three systems for the Rastrigin function with $K = 4$ and varying D , N and α in Table 1. We evaluate the error $\mathbb{E}[\|\bar{x}_T^N - x^*\|]$ and the success rate. Both are estimated via a Monte-Carlo approximation with 10000 runs, where the expectation denotes the average deviation across all runs and the success rate is defined as the proportion of runs satisfying $\|\bar{x}_T^N - x^*\| < 10^{-2}$.

D	α	measure	$N = 2^5$			$N = 2^8$		
			QMC ² BO	CBO _↓	CBO _↑	QMC ² BO	CBO _↓	CBO _↑
2	1	success rate	0.00%	4.08%	4.92%	0.00%	23.28%	18.92%
		error	2.86e-2	2.22e-1	3.07e-1	1.14e-2	1.81e-2	2.05e-2
	10	success rate	100.00%	33.52%	45.31%	100.00%	96.64%	98.39%
		error	8.18e-3	7.97e-2	8.55e-2	3.42e-3	4.19e-3	4.21e-3
	100	success rate	100.00%	82.09%	90.86%	100.00%	100.00%	100.00%
		error	2.67e-3	6.57e-2	7.81e-2	1.40e-3	1.47e-3	1.40e-3
3	1	success rate	0.00%	0.08%	0.22%	0.00%	1.87%	1.52%
		error	9.66e-1	7.78e-1	8.23e-1	4.32e-2	1.04e-1	1.63e-1
	10	success rate	0.00%	2.29%	6.38%	100.00%	27.48%	45.92%
		error	1.72	5.67e-1	4.83e-1	1.43e-2	1.68e-2	1.08e-2
	100	success rate	0.00%	18.65%	45.06%	0.00%	92.59%	99.97%
		error	1.72	5.38e-1	4.64e-1	9.94e-1	6.65e-3	3.83e-3

Table 1: Monte-Carlo approximation (10000 runs) of the success rate and the error $\mathbb{E}[\|\bar{x}_T^N - x^*\|]$ for the QMC²BO method 2.7 and the CBO method (1.2)-(1.3) with $\sigma = 1$ (CBO_↓) and $\sigma = 1.75$ (CBO_↑), applied to the Rastrigin function for varying N , α , and D .

For all three systems, the error and success rate improve with increasing N and α , and worsen with increasing dimension D . An exception occurs for QMC²BO at $D = 3$, where $\alpha = 10$ yields a better approximation than $\alpha = 100$.

More interestingly, the deterministic and stochastic systems differ in their behavior. The outcome of the deterministic QMC²BO method is identical across runs, resulting in a complete success (100%) or complete failure (0%). The stochastic CBO variants rarely attain these extreme values. For example, in $D = 2$ with $N = 2^5$ and $\alpha \in \{10, 100\}$, QMC²BO consistently achieves a 100% success rate, while stochastic CBO reaches only about 80%-90% or even 30%-45%, depending on α . In contrast, in $D = 3$ with the same parameters, the deterministic method fails, while stochastic CBO still achieves moderate success rates.

The main takeaway is that stochasticity can be both a benefit and a drawback. It can encourage convergence to the minimizer in situations where the deterministic method fails, but it does not behave consistently across all realizations.

5 Conclusion

We studied the QMC²BO method 2.7, a zero-noise variant of CBO initialized by low-discrepancy Quasi-Monte Carlo points. Under mild regularity assumptions on f , we showed that the particle average $\bar{x}_t^N$ converges to the global minimizer over time. In Theorem 2.8, we derived explicit bounds for $\|\bar{x}_t^N - x^*\|$ in terms of the number of particles N and the concentration parameter α . We quantified how to choose N , α , λ and the time horizon T to achieve a prescribed accuracy $\varepsilon > 0$.

The error decomposition in Theorem 2.8 revealed three contributions. The term $e^{-\lambda t} \|\bar{x}_0^N - x^*\|$, which decays exponentially. Without prior knowledge of x^* , this contribution cannot be reduced. The

remaining contributions are captured by $\mathbf{C}_{\text{Lap}}(\alpha)$, arising from the Laplace principle and the use of finite α , and $\mathbf{C}_{\text{QMC}}(N, \alpha)$, reflecting the Quasi-Monte Carlo approximation error due to finite N .

A central part of the analysis in Section 2 was about deriving explicit and computable bounds $\mathbf{C}_{\text{Lap}}(\alpha)$ and $\mathbf{C}_{\text{QMC}}(N, \alpha)$. Our focus lies on obtaining transparent and tractable estimates, not on deriving the sharpest bounds possible. Hence, our results should be understood as a first step towards combining deterministic CBO dynamics with Quasi-Monte Carlo theory to obtain explicit convergence rates. There is potential for refinement, both by taking advantage of more advanced results from Quasi-Monte Carlo theory to sharpen $\mathbf{C}_{\text{QMC}}(N, \alpha)$ and by improving the estimates in Section 2.2 to sharpen $\mathbf{C}_{\text{Lap}}(\alpha)$.

The numerical experiments in Section 4 confirmed that QMC²BO reliably converges to the minimizer for several benchmark functions with different properties, including functions beyond the assumptions of Theorem 2.8. The observed dynamics consists of two phases, an initial coordination phase dominated by finite- N and finite- α effects, followed by a phase where the exponentially decaying term dominates. Some functions show both phases; others skip the coordination phase and jump directly to exponential convergence.

A comparison with stochastic CBO highlighted the role of randomness. Noise enables particles to leave the convex hull of the initial configuration and can enable convergence in cases where the QMC²BO method fails. At the same time, stochasticity introduces variability and can make the method less reliable.

Acknowledgments

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